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Kenneth L. Cheney · Expound Laboratories™ · August 2026

> **Third in a series about friction.** *Who Can Afford Friction?* took the general case; *Startup IP Is the Most Underpriced Asset on Your Cap Table* took patents. **This one is about the place the price has not moved** — and the next section makes all of it self-contained.

First, some words you will need

Before the argument, the vocabulary it runs on. Eleven entries, each defined once, in plain English. If you already know them, skip to *What this is about* below — nothing here is specialist, and none of it is jargon for its own sake.

Your AI vendor. Throughout this article, the party your company buys AI capability *from*, and the counterparty to every clause discussed below. That is a wider set than it sounds: the foundation-model provider you call directly, the cloud platform reselling somebody else's model, the SaaS application with a model embedded in it, and the systems integrator operating any of them on your behalf. Each of those is a separate agreement, and **for each path your material actually travels, your exposure is bounded by the least protective applicable agreement, configuration and downstream processor on that path.**

Inference. The system reads your question and produces an answer. In the narrow sense, the model's own parameters do not change.

Training. The system's parameters change because of something it saw. Something persists, and the next user gets an altered system.

Weights. The numbers inside a model that determine how it behaves. Training changes them.

Fine-tuning and adapters. Cheaper, smaller ways of changing behavior without retraining everything.

Parametric, and non-parametric. *Parametric* means the change lives in the model's own numbers — the weights. *Non-parametric* means it lives beside the model instead: in a stored note, a searchable index, a saved example. The distinction matters because the published protections I have reviewed are far more explicit about parametric training than about durable non-parametric state.

Traces and evals. A *trace* is the record of what the system did on a task. An *eval* is a test you wrote to check it is getting your work right. Both are things your organization creates.

Corpus and checkpoint. A *corpus* is the body of material a training run was given. A *checkpoint* is the saved state of the model at a point in that run — the artifact that gets released as a product. Tying a released checkpoint to the corpus behind it is the thing most verification proposals are trying to do.

Attested execution. A pipeline that produces cryptographically verifiable evidence about the measured software and configuration that ran, under declared hardware and trust-root assumptions, so the claim does not rest solely on the operator's word. It does not prove that the software behaved as intended or identify the corpus it consumed, and it does not establish that the code does what its author says.

Tenant. Your walled-off area inside a vendor's cloud. It does not mean the building is yours.

Exclusion right. A right to stop someone doing something — the power to say no. **Exclusion is the economically interesting incident of property, but a contractual restriction on use does not by itself create a property right**: contract, confidentiality, privacy and IP law all contain exclusionary incidents without making every enforceable veto property. This article argues that more of that kind of veto may already be available to you than you think — depending on your agreement, the data involved, the parties' legal roles and the governing law.

One note on direction. You are the *customer* in everything that follows. The mirror-image exercise — what your own agreements let you learn from your customers' use of your product — is this same list read from the other side, and the clause list at the end works in that direction too.

What this is about, in plain terms

You bought an AI system — an assistant your teams work in daily, a copilot inside your helpdesk or your CRM, a model your engineers call directly. To make it useful, your people spent months telling it how your business actually works — which answers are wrong and why, what your terms mean in practice, where the exceptions are that nobody ever wrote down. That correction work cost real money in salaried hours. **It appears on no invoice.**

In July the chief executive of Microsoft gave this a name: the Reverse Information Paradox. His claim is that you pay for intelligence twice — once in cash, and once in the knowledge you had to hand over to make the thing work at all.

Here is the part that goes largely unsaid. What you cannot buy routinely is a cheap, independent, customer-level answer to whether that second payment was collected. The standards do not supply one. Litigation is a route to stronger evidence and an expensive one. And the techniques available to you from outside your vendor's infrastructure are partial, contestable, or both.

That is the subject of this article — not whether you are being charged twice, but whether you could ever establish it. And the answer turns out to be more useful than the question, because a thing you cannot afford to check is a very different problem from a thing you have no right to.

It also produces something you can use tomorrow, so here it is up front rather than at the end. One question comes first, and then four properties decide whether a record can be checked by someone who does not trust the party it describes. **Nothing about them is specific to AI** — they apply to any record handed to you by any company you buy from, about anything that company says it did:

> **0.** *First, scope:* what proposition is this record meant to establish, does its declared scope cover what that proposition needs — and who had authority to decide both?
>
> **1.** The outcome the record establishes cannot be set by the party it describes.
>
> **2.** What had to be true is fixed in advance — with the "before" evidenced by a time or ordering authority the subject cannot unilaterally set — and the subject cannot narrow it after adoption.
>
> **3.** Evidence is graded by what it can actually support; the mapping is fixed before composition under authority the subject cannot unilaterally alter, and classification is governed too.

> **4.** The decision behind the record reconstructs from the outside — for somebody who was not there and does not trust you.

Question zero is not one of the four, and it comes first — a perfectly checkable report about the wrong system is a perfectly checkable report about the wrong system. **Drop any of the four and you may still have evidence; what you no longer have is the form of independent verification defined here.** The reasoning behind each — and what you are accepting when one is missing — is in *Four properties that make a properly scoped determination checkable without trusting its subject*, below.

One distinction to carry with you

Before the argument starts, one piece of care that everything else depends on.

Over those months of correction, the system you bought got better at your work. **The natural way to say that is *your people taught it* — and that is already more than anyone outside the vendor can observe.** What you can see is that corrections went in and the output improved. What the improvement rests on — a longer prompt assembled for this session, a memory record kept in your own area of the vendor's cloud, an example added to a test set, a change to the model itself, or nothing at all that outlives the conversation — differs by product, and in the documentation I have read most vendors do not say which.

So keep two things apart. The corrections are yours and you can see them. What became of them, you cannot.

The essay Nadella published in July states the strong version throughout, in the register that makes it feel settled: that models learn from "exhaust," that "every correction is distilled into institutional know-how," that it "leaks almost imperceptibly: trace by trace, correction by correction, eval by eval." **Those are his sentences, quoted in full later in this article, and I quote them rather than adopt them.** They may well be right. They are not checkable from where you are standing, and the difference between *probably true* and *established* is the entire subject here.

Nadella asks what instrument would keep that improvement with the firm that produced it, and offers patents as the precedent. He also gives his own answer, which is not legal at all: **distribute the learning infrastructure to every firm, so each controls its own learning loop** — the whole cycle by which a system is corrected, retrained and redeployed on your own material.

My answer is that no legal instrument owns it, and that what you can have instead is narrower than a new right, and may already be available under existing law and contract — depending on

the jurisdiction, the facts, the data, the parties' legal roles, the configuration and the agreement, and provided you go and get it. The heart of this article is the four properties above — what any record has to satisfy before someone who does not trust its subject can check it. They are the durable part, they belong to no vendor, and they outlast this argument. **The instrument that puts them to work is a list of things to raise with the company that supplies your AI** — the model provider, the cloud platform reselling one, or the software vendor with a model inside it. The properties are what you require. The clause list is where you require it.

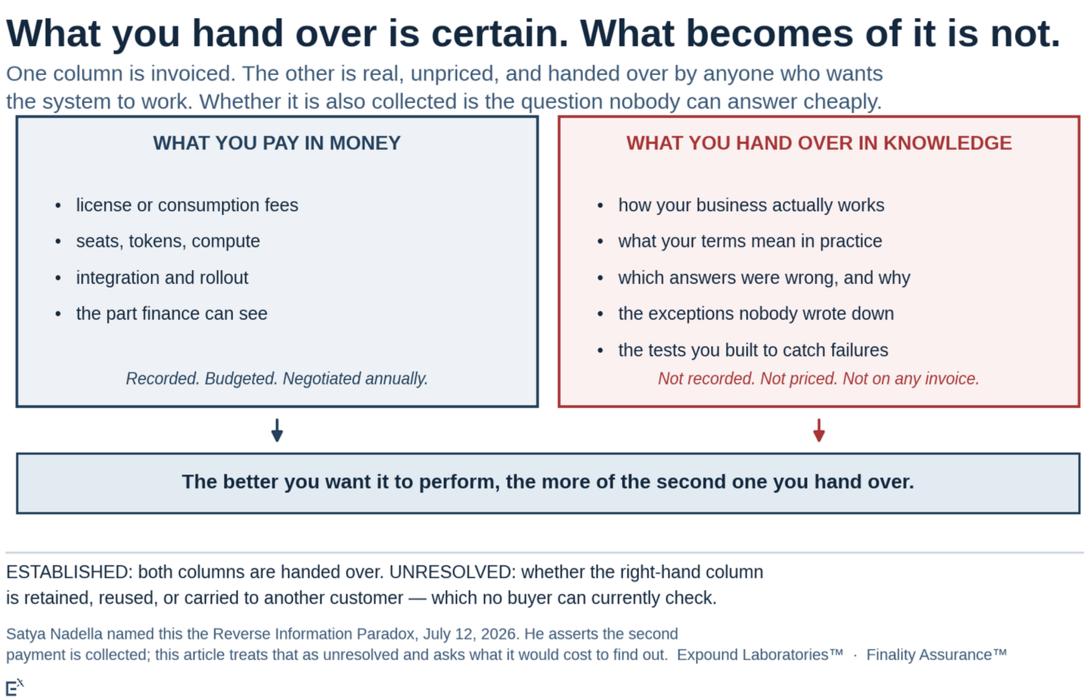

Figure 1 *What you hand over is certain. What becomes of it is not.*

The whole argument in one sentence

Before any of the law or the machinery, here is the shape of it.

You are short one standardized entitlement, in one place. Everywhere — that place included — you are also short an affordable way to find out.

Everything in this article is one thing wearing different clothes: a contract carve-out you never read, a standards vocabulary with no word for your problem, an attribution study you cannot afford to run, and a mathematical guarantee I have not found offered as standard. Each gets its own section below. **Only the first is a gap in your entitlement. Most** are places where the cost of establishing what is true is higher than the value of knowing it — **attribution is the important mixed case, where depending on the proposition and the granularity the limitation may be economic or methodological** — so nobody finds out, and the not-finding-out becomes the status quo.

What friction means here, and how to test for it

The word carries this article, so it is worth pinning down rather than gesturing at. **You do not need the earlier articles for any of this.**

> **Friction is the cost of establishing where you stand — and it is not the same thing as a bar on standing there.**

Most discussions of rights ask one question: *am I allowed?* When the answer turns out to be "you never found out," there are three different reasons why — and it is easy to collapse them into one.

The first is a bar. You are not allowed. The answer is "no," it is written down somewhere, and money does not move it. **Fixing a bar requires a legislature, a regulator, or a court** — someone with authority to change what the rule says.

The second is a price. You are allowed. Nobody disputes it. But finding out where you actually stand — reading the agreement, getting the opinion, running the analysis, proving the fact — costs more than the answer is worth to you. So you do not find out. **That is friction, and fixing it requires somebody to build a cheaper instrument** — engineering and procurement rather than politics.

The third is a technical gap. The answer does not exist to be had, at any budget, because nobody knows how to produce it yet. **Fixing that requires an invention**, and pretending it is a price is how people waste years.

So the test is two questions, in order. Is there a rule that forbids it? If yes, it is a bar. **If no: would the answer exist, if somebody built the instrument and you paid for it?** If yes, it is friction. If no, it is a technical gap, and the honest thing is to say so rather than call it a price.

That second question is the one that does the work, because the *experience* of all three is identical — you end up not knowing, and act as though you had no right at all. **That identical**

feeling is why the distinction gets missed, and missing it sends people to the wrong remedy for years.

Two hard assurance problems remain, and neither is an unknown technical primitive. Establishing that two evidence sources fail *independently* rather than merely sitting in different reporting lines carries residual epistemic uncertainty. And **end-to-end corpus completeness is an unstandardized engineering and assurance problem**: known mechanisms can bind a declared manifest to an attested execution, but whether that manifest faithfully captures every transformation, every generated input, the authorization state of each source and the resulting checkpoint is the harder question. **Almost everything else here is a price.**

Three properties follow, and each does work later in this article.

One. Friction is invisible, because nobody ever decides to accept it. A bar produces a refusal you remember. A price produces a meeting that does not get scheduled, a clause nobody reads, an analysis nobody commissions. **There is no moment at which you gave the right up**, which is why a right can go unexercised across an entire industry without a single person choosing that outcome.

Two. Friction always has a beneficiary. Somebody is spared an argument by the fact that checking is expensive. That party has no obligation to reduce the cost and usually no incentive to. It is worth asking, at every point below, who that is.

Three. Friction is reversible, and on a much shorter timetable than law. A bar waits on an institution. A price waits on an instrument — and prices move when somebody builds the cheaper thing and enough buyers ask for it. **This is the good news, and it is why the distinction is worth this much space.**

Two prior instances, told so you do not have to go and read them. *Who Can Afford Friction?* took the general case: what stops people acting on rights they already hold is rarely the law, it is the cost of finding out where they stand. *Startup IP Is the Most Underpriced Asset on Your Cap Table* took a specific one — for patents, the cost of establishing what a portfolio contains and what it is worth collapsed, and the asset repriced without a word of law changing. **Same right, different price, different world.**

> **The cost of discovering what your vendor did with what you gave it is still an institutional cost. That is friction, not law — and it is the last place in this series where the price of knowing has not moved.**

So read the rest of this article with the test in hand. Four times below, the answer will look like a prohibition and turn out to be a price. Those four are marked.

What I am claiming, and what I am not

An article about unverifiable claims should be explicit about the status of its own. Three categories, held to throughout — established, assumed, hypothesis — plus one list of what I am deliberately keeping out of the first.

Established — and I am drawing this line tighter than such lists usually are. Something counts here only if it sits on *your* side of the boundary: a thing you can observe directly, or a document you hold. A vendor's description of its own internals establishes the representation the vendor made. **It does not, without more, independently establish that the system behaved as represented** — which is the same standard the article later applies to a vendor-generated log.

- You reveal proprietary operational knowledge to make a purchased system useful. A description of how the systems are used, not an allegation about anybody.
- **Some retention is directly observable to you.** Conversation history you can scroll back through, a memory feature that recalls last month's correction, an admin console listing stored transcripts, a retrieval index you populated and can query. You do not need anyone's word for these.
- **Your own agreement says what it says.** Whatever it permits beyond answering your request is readable in a document sitting on your side of the table.
- The published standards vocabularies have no *machine-readable* category for this channel — the one field that could carry it is a free-text string no system parses, attached to a file rather than to a trace. Readable in the specifications, cited below.

And here is what I am deliberately not putting in that list, though most writing on this subject does.

- **Retention you cannot observe** — telemetry, trace logs, evaluation corpora, reward signals, anything held in infrastructure that surfaces nothing to you. Vendors document a good deal of this, and product documentation is the natural thing to cite. **It is also a vendor statement about vendor internals.** I am not calling it false; I am declining to call it established, because accepting it here while rejecting a vendor-generated log two sections later would be having it both ways.
- **That a broad product-improvement carve-out is present in every enterprise agreement in the sample I reviewed.** That is a complete statement about my sample and none at all about yours. This is my reading of the agreements I have seen, and it matches what practitioners report. It is an observation across a sample, not a fact about yours. **Go and read yours** —

which is what the preamble to the clause list asks you to do first, and the reason it comes before item 1.

Assumed — load-bearing, stated so you can reject them.

- That enterprises would act differently if checking were cheap. Plausible, and unmeasured.
- That the absence of a verification offering reflects coordination failure rather than absence of demand. Argued below, not demonstrated.

Hypothesis — asserted by others, unresolved by me.

- That the retention amounts to a **second payment**: that your contributed experience measurably improves a system your competitor then buys. Note the split, because it is where most writing on this subject goes wrong. *Observable retention* is established above, and so is *whatever your own agreement permits.* What is unresolved is whether the value **travels beyond the vendor itself** — whether learning derived from your contribution becomes available to another customer or a competitor. **Nadella does argue that the seller learns from the buyer, and that value can converge toward the owner of the learning infrastructure. He does not establish the stronger proposition, and I do not adopt it as fact. I want to be exact about why.** From outside a vendor's infrastructure that claim is not testable at acceptable cost, which means I can neither support it nor refute it, and neither can anyone else making confident noise in either direction.

What this article resolves.

Not whether you are being charged twice. **Whether you could find out.** And the answer to that is specific, defensible, and unhappy: not cheaply, not without litigation, not through the standards, and not, so far as I have found, from anyone who does not already trust the vendor.

> **The vendors are not necessarily doing anything wrong. The defect is that "necessarily" is doing all the work in that sentence, and nothing I have found available to you removes it.**

That is a narrower claim than the one usually made about this subject. It is also the one I can defend.

Before you read on

Two audiences, and I am not saying the same thing to each.

If you buy or run AI inside a company:

1. **You are handing over something valuable that no invoice records** — the proprietary knowledge your people must keep revealing to make purchased intelligence work. Some of the retention you can see for yourself, and your own agreement says what it permits. Retention you *cannot* see rests on your vendor's description of its own systems — which is a representation, not a fact you have checked.
2. **Your existing protections are better than you think and narrower than you were told.** Trade secret law already reaches misuse without needing to prove anything was copied — but only to the extent your own contract has not already granted the use. The carve-outs you never read are not merely the places a leak could run; they may later give the vendor a contractual argument that the challenged use was authorized.
3. **What remains is an unstandardized entitlement in one place and a verification gap throughout.** The layer holding memory records, retrieval indexes and evaluation sets — defined in the next section — has the least standardized entitlement structure *and* is unverifiable. Elsewhere you are probably entitled and still cannot check.

If you build or sell AI:

1. **The distinction between using your system and learning from your customer is asserted in the agreements you issue and absent from the standards.** The published vocabularies have no category for it, which means your own customer terms are doing all the work.
2. **Ask whether the record you keep would be evidence to a party who does not trust you.** That difference is where the commercial opportunity sits.
3. **Hardware attestation turns some execution claims from an operator's assertion into cryptographically verifiable evidence** about the measured software and configuration that ran, under a declared hardware and trust-root model — evidence that rests on the chip rather than on the operator's word. **On its own it does not establish what went into a training corpus**; that needs a chain tying data admission, corpus identity, preprocessing, job and checkpoint to the attested run. The obtainable object is narrower and more useful: controls that prevent your material from being admitted at all, and evidence sufficient to determine afterwards whether it was. **Differential privacy** adds mathematical noise during training to bound, under stated parameters and a defined privacy unit, how much the distribution of a trained mechanism's outputs may differ between adjacent datasets that differ in whether that unit participated — at a real cost in accuracy. Note the shape of that guarantee: it bounds a *ratio*, under parameters somebody chooses, and depending on ε, δ, the privacy unit and cumulative composition the resulting bound may remain far looser than the assurance a customer actually wants. **Neither requires a new right. Attested execution is available today.** What I have not found offered as a standard frontier-provider assurance is the customer-level binding from an attested learning process to the identity and authorization state of the

material it consumed, or per-customer differential-privacy accounting where learning is authorized.

The correction that matters

And now the correction that matters, because the popular version of this subject gets it wrong.

"Inference changes nothing" is true of the model and misleading about the system — and in most enterprise products you can see the difference. The narrow claim is the one just defined, and it is correct: answering your question does not update the weights. What the shorthand hides is that a single request visibly writes a memory record you can later recall and an index entry you can query. Vendors additionally describe feedback events, traces and routing signals; those are their account of their own systems rather than something you have checked. **Either way, none of it is training**, which is the point.

So the useful distinction is not two states but **three levels of persistence**:

Level	What it is	Does it persist?
Ephemeral	state that does not survive the request or task boundary — transient activations, strictly task-scoped working context	No — gone when the interaction ends
Durable but non-parametric	state outside the model's learned parameters that survives the request *for any period*: memory records, search indexes, saved examples, prompts, routing preferences, eval sets, traces, annotations — **and short-lived session stores and prompt caches**	**Yes**, and it is the least governed
Parametric	weights, fine-tunes, adapters	**Yes**, and it is the one everyone argues about

The line between the first two rows is not long retention versus short. It is whether state survives the interaction, and where it lives. A cache that holds your inputs, outputs and derived data for twenty-four hours is not ephemeral in any sense that matters to this argument — it is durable non-parametric state with a short clock on it, and the clock is a vendor configuration rather than a property of the technology. Reading it the other way makes the middle row look smaller than it is, and the middle row is the whole problem.

Read that table as a map of what *can* persist, not a finding about your vendor. The middle row is where products advertise features — memory, history, retrieval over your corpus — so some of it you can see working. The rest of it is the architecture as vendors describe it. **A taxonomy of possible channels is useful precisely because you cannot audit which ones are active**, and it is worth being clear that this is a frame for asking questions rather than an inventory of what is happening to you.

Almost all the public argument is about the third row. The second row is where the governance is thinnest — and persistence is not the same thing as escape, so it is worth being precise about how the middle row actually crosses a boundary.

Within your own tenant, a memory record that persists forever may never reach anyone else. **Durable state crosses the customer boundary through several routes, including:**

- **Retrieval corpora assembled across customers**, where an index built partly from your material would serve other accounts.
- **Eval sets promoted into product benchmarks**, which is how the tests you wrote to catch your failure modes would become the vendor's regression suite.
- **Routing and model-selection policies updated from aggregate feedback**, where your accept and reject signals would tune a policy every customer is served by. This is the route with the least obvious query surface: a retrieval index can be probed and a system prompt can often be extracted, but a routing policy is weights inside a component you cannot address, and its behavior is visible to you only as which model answered.
- **Prompts and few-shot exemplars carried into system prompts** — the cheapest and least logged of these, though also, because they sit in the serving path, among the more extractable once you go looking.
- **Shared safety, moderation and classifier development**, where your material improves a model every customer is screened by.
- **Synthetic-data generation from traces**, which produces training material carrying no marking back to its source.
- **Human review and annotation** — a person reading your prompts and completions. The lowest-technology route on this list, and it needs no model at all.
- **Support, debugging and incident access** to tenant data.
- **Cross-tenant or shared caches**, where applicable.

That list is representative rather than closed, which is itself the point: a taxonomy you can extend is a taxonomy that will need extending, and a clause list built on a closed one will govern the routes somebody thought of in 2026.

And one thing on that list is not a route at all. The **"aggregated or de-identified product improvement" carve-out** is not another technical mechanism by which state crosses a boundary. It is contractual permission that may authorize one or more of the routes above — which is why it does more work than any of them.

That last one is the point. The parametric row is shared by construction and therefore argued about. **In many enterprise arrangements** customer-specific persistence and broader vendor use are treated differently, with the broader use depending on specified exceptions or separate permissions — and those exceptions live in a clause that is easy to overlook. **Read your agreement rather than assuming that structure applies to it.**

> **Friction, instance one.** Nothing prevents you reading that clause. It is in the agreement you signed. What stops you is that finding it, understanding what it licenses, and pricing what it costs you takes counsel time nobody budgeted against a harm nobody has quantified. The clause is not hidden. It is merely more expensive to evaluate than to accept.

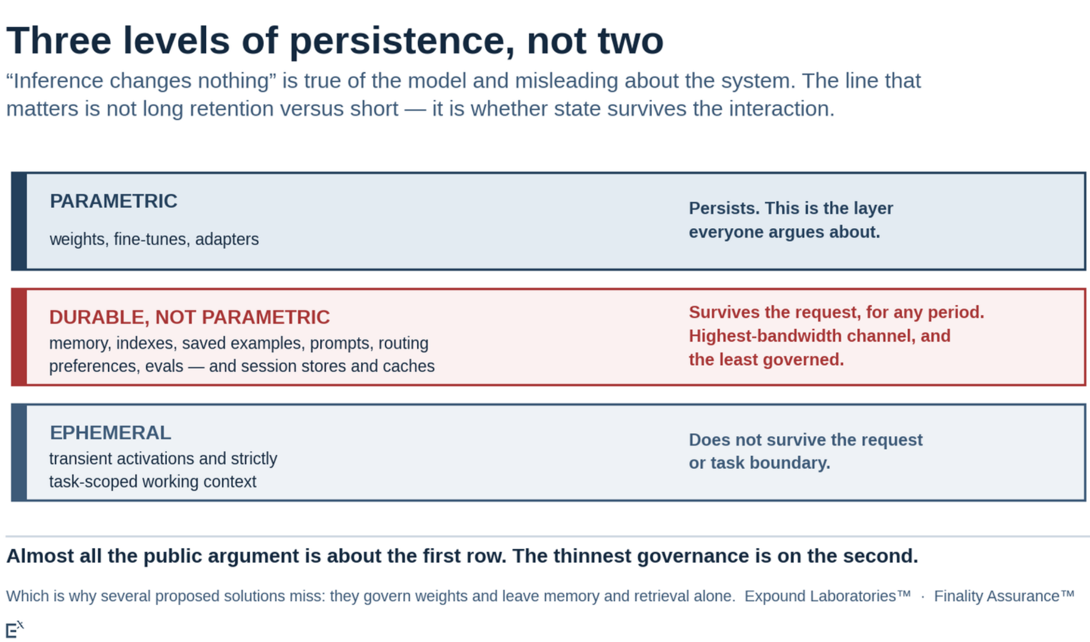

Figure 2 *Three levels of persistence, not two*

What Nadella actually said

On **July 12, 2026**, published under his own name rather than through any Microsoft channel, Satya Nadella posted a long-form essay subtitled *"In the age of intelligence, how should firms protect their core IP?"*

He begins with a famous problem described by Kenneth Arrow — later a Nobel laureate — in 1962:

> "Nobel Prize winning economist Kenneth Arrow famously described a paradox in the market for information. 'Its value for the purchaser is not known until he has the information, but then he has in effect acquired it without cost.' In Arrow's 'Information Paradox,' the seller risks giving away knowledge in order to sell it."

In plain terms: to sell you an idea I must show it to you, and once I have, you already have it. That is why selling ideas is hard, and it is one information problem patent rights can help address.

Then he turns it over:

> "AI creates the reverse problem. In the AI age, the buyer risks giving away knowledge, just in order to use what they bought."

>

> "You essentially pay for intelligence twice, once with money, and again with something even more valuable: the proprietary knowledge you must reveal to make that intelligence useful. The better you want the model to perform, the more of that knowledge you have to feed it!"

And the open request, emphasis mine:

> "Patents solve one aspect of Arrow's paradox. They let an inventor disclose an idea without simply giving it away. **The Reverse Information Paradox needs its own equivalent.**"

Then the sentence that makes it more than an observation:

> "This requires more than data protection. Models learn from 'exhaust,' the prompts people write, the tools agents use, and especially the corrections people make when

> the model is wrong. Every correction is distilled into institutional know-how. It's the kind of knowledge a competitor could never buy, and the kind that leaks almost imperceptibly: trace by trace, correction by correction, eval by eval."

And he answers his own question, which the coverage I have read omits. The essay's operative proposal is architectural rather than legal:

> "Therefore, it's imperative that we distribute the learning infrastructure to every firm so that they can control their own learning loop."

Alongside it he asks firms to draw a **trust boundary** around organizational data, traces, evaluations, adapted model weights and memory — and, separately, the mechanisms through which the firm learns — keeping them under enterprise control unless explicitly shared. Hold that proposal; two sections from here it gets the scrutiny it deserves.

Two corrections to his framing, and both make the problem more tractable rather than less.

First, the patent analogy illuminates the disclosure problem and points to the wrong legal object for most of the knowledge at issue. A patent requires you to publish an enabling description of your invention, and patentability turns on eligible subject matter, novelty, non-obviousness, utility and adequate disclosure. The knowledge your people hand over — that a certain clause always signals a certain risk, that this customer segment behaves differently — is accumulated operational judgment, valuable only while it stays unpublished, and it is not the kind of technical invention for which patenting is the practical protection mechanism. **For the confidential know-how at the center of Nadella's example, trade-secret law is generally the closer fit**, alongside contract, confidentiality obligations and data-protection law. That is not an exclusive claim: technical implementations of the machinery for governing learning may well be patentable, and are a different object from the business judgment being contributed. What changes is which regime does the work here. More on this below.

Second, this is not merely Arrow's original problem, and the difference matters. Arrow's paradox is about *valuation under uncertainty*: you cannot price what you have not seen. **The buyer may well face Arrow-like uncertainty about the AI capability it is purchasing** — its quality, its suitability, how it will perform on their work — as buyers do with many complex technologies, and I am not going to pretend otherwise to make a cleaner contrast. **Nadella's distinctive problem begins after the purchase**, when making that capability useful requires a relationship-specific investment of proprietary know-how. At that point there is a second economic object in play, and **holdup and spillover explain something about it that Arrow alone does not.**

What they have is two failures wearing one name.

The first is holdup. Teaching a vendor's system how your business works is an investment that becomes progressively harder to move, and once it is made your bargaining position on renewal is worse than it was. That is Williamson and Klein-Crawford-Alchian territory, and it comes with a developed remedy set — commitment devices, hostages, audit rights, exclusivity, and vertical integration as the last resort.

The second is spillover, and it is the one that stings. The knowledge is valuable *outside* the relationship too — so if it does travel, the beneficiary is your competitor rather than your vendor. That makes it an appropriation problem rather than a pure bilateral one, and it is the half the holdup literature handles least well.

I flag this because it constrains the answer. If it were only holdup, contract remedies would be sufficient and this article would be shorter. It is the spillover half that makes verification the load-bearing requirement: a remedy that binds only your counterparty does not help if the harm arrives through a third party who never signed anything.

The name is his and it has stuck, so I will keep using it.

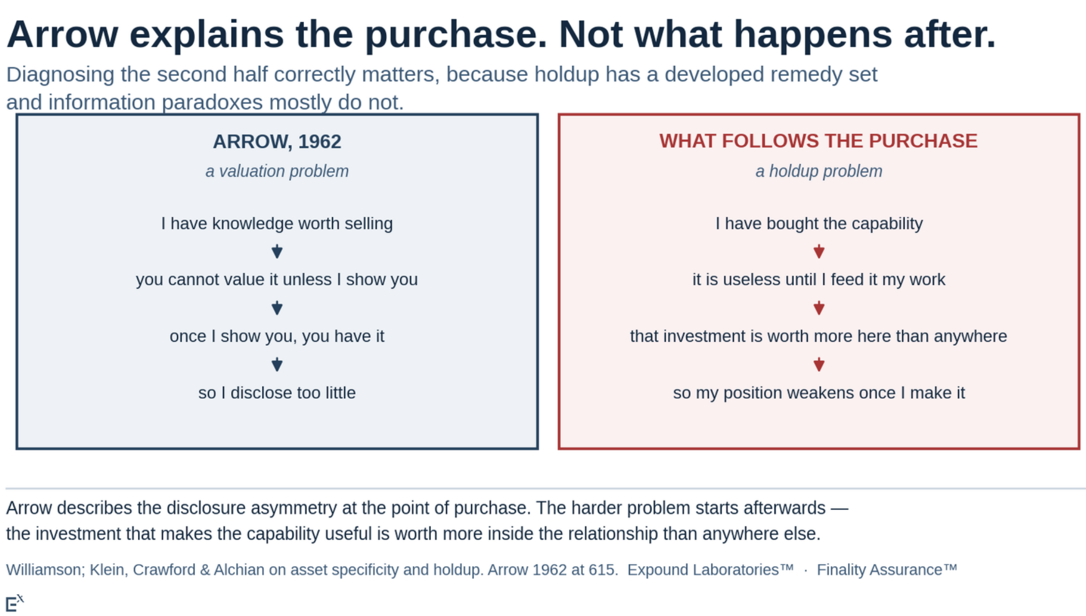

Figure 3 *Arrow explains the purchase. Not what happens after.*

Microsoft's commercial position, and why it changes how you read the prescription

Before I examine anyone else's incentives, symmetry requires me to disclose my own. I build systems intended to make this class of determination governable and independently checkable, so I have a commercial interest in the problem being taken seriously. The specific systems, what they do and do not establish, and the validation posture behind them are set out in full later in this article — but you should have that fact before you read the next few paragraphs, not after.

Microsoft sells products and services responsive to the problem he describes. The Register ran the essay as *"Lock it down, warns Satya Nadella, seemingly forgetting the billions Redmond chipped in to OpenAI back in the good old days."* The Next Web: *"Nadella says you pay for AI twice, and Microsoft helped build the trap."*

All legitimate; none of it disposes of the argument. **A seller naming a real problem they also monetize is the ordinary condition of every industry.** And to his credit, he turned the same criticism on his own side, after first endorsing fair-use rights for model providers to train on public data:

> "While the great innovation that comes from model providers having fair use rights to train models on public data is needed, I find it ironic that the status quo is to then turn around and impose restrictive terms on distillation, and to reserve the right to learn from customer usage and interaction data. If learning flows in only one direction, economic value converges toward the owners of the learning infrastructure rather than the creators of the knowledge itself."

Distillation there means using one model's outputs to train another — forbidden in many vendor terms while those same terms reserve the right to learn from you. **That asymmetry is the whole argument in one sentence, and it is contractual, which means it is negotiable.** Nobody legislated it. Somebody drafted it.

But "Microsoft sells the fix" is too blunt an answer, and worth doing properly — because motive shapes which parts of the essay you should lean on and which you should discount. Applying the same discipline this article applies to everything else: one of the following is established, the rest are inference.

On the public record — a looser standard than the claims ledger above, since press reporting is not on your side of the boundary. Adequate for reading an author's motive; inadequate for anything about what a system does. Ten days before the essay, on **July 2, 2026**, Microsoft announced **Microsoft Frontier Company**: a new operating business with a **$2.5 billion** commitment and **6,000 industry and engineering experts**, created to go inside enterprises and deliver AI deployments. Judson Althoff, Microsoft's Commercial Business CEO, described it as going "beyond what has been labeled as Forward-Deployed Engineering" and as "the largest, most capable, outcome-driven engineering organization in the industry." Early named engagements included the London Stock Exchange Group, Unilever and Land O'Lakes, with Accenture named as a delivery partner.

Also on the record. Asked directly about the essay, a Microsoft spokesperson identified Copilot and Azure AI Foundry as the company's answer to the problem it describes. That is not a critic's inference; it is the vendor naming its own product as the remedy.

Also on the record: it was not a solo move. AWS committed $1 billion to its own deployment organization on June 30, two days earlier. OpenAI and Anthropic had launched comparable ventures that May.

Everything past this point is my reading, and the dates do not compel it. What the sequence suggests is that the supply was funded first and the demand articulated after — a capability staffed on July 2, then an essay on July 12 explaining in economic language why every enterprise needs the thing that had just been staffed.

The innocent explanation is available and I should say so. Large companies announce operating units and publish executive essays in the same month routinely; fiscal-year timing clusters both; an essay drafted over months can land near an unrelated launch. Ten days is suggestive, not probative, and I have no evidence about drafting order.

And the strongest discount on my reading is a matter of public record, in the looser sense flagged above: three competitors funded the same kind of organization inside ten weeks. An industry-wide move to buy deployment capacity does not need an essay to manufacture demand for it. **If you want to discount the motive reading entirely, the argument of this article does not change** — which is the test I would apply to anyone else's inference about motive.

The commercial alignment is worth stating once, plainly, rather than enumerated at length. Every remedy the essay proposes runs on infrastructure: keeping data, evaluations, memory and adapted weights inside a boundary you control describes a cloud tenancy with governance tooling attached, which is a product Microsoft sells. The essay does not resolve the adjacent Copilot question: whether using customer information at inference time to retrieve, reason over or personalize a response is economically or legally different from admitting that information into a shared learning process. That distinction matters to everything below, but it is mine — he does not draw it in the essay. What the essay does do is raise the salience of learning terms, which favors whichever vendor believes its terms are tightest. **And naming a problem is how you come to own the vocabulary for it**: "Reverse Information Paradox" is the name every subsequent piece I have seen has used, and it frames the answer as *control your loop* rather than *audit your supplier*.

None of that makes the diagnosis false, and I am not going to spend more of this article on it. It is inference about motive, it is the easiest thing in this piece to argue about, and arguing about it is a detour around the question that actually matters.

And the least cynical reading, which I think is also partly true. He published under his own name rather than through a Microsoft channel, argued in economics rather than in product, and included a criticism that lands on his own industry and to some degree on his own company. That is not the shape of pure marketing. **The honest position is that all of these can be true at once**, and that a chief executive of a platform company has no available register in which to think aloud about platform economics without also doing commercial work.

Why this matters for your reading, concretely. Take the diagnosis seriously and the prescription skeptically. The description of the second payment costs him nothing and is the strongest part of the essay. The remedy — own your learning loop — happens to require buying more infrastructure, and is the part where a reader's interests and the author's diverge most. **That divergence is exactly where I part company with him**, and the next section is why.

Nadella's own answer, and why it is right and expensive

Return to the proposal quoted earlier: distribute the learning infrastructure, and draw a trust boundary around data, traces, evaluations, adapted weights and memory.

Two things follow, and the first is a concession. That list — data, traces, evals, adapted weights, memory — is the durable non-parametric layer plus the parametric one. He has the taxonomy already. Anyone claiming the three-level distinction as novel should read his paragraph again.

The second is where I disagree. Moving the learning loop inside a boundary you control is a move toward vertical integration, and at the strong end it becomes vertical integration outright. In the holdup literature it is the remedy of last resort, chosen when contracting has failed, and it is chosen because it is expensive — you take on the capital cost, the talent, the infrastructure and the obsolescence risk that the specialization was supposed to spare you. **And it is worth being precise about who can actually do it**, because the answer explains the entire wave of **forward-deployed engineering** — vendors placing their own engineers inside customer organizations to build the thing for them. Owning a learning loop end to end requires five things at once: model weights you are licensed to *train* rather than merely call; compute at training scale; platform engineers who can run the pipeline; **an evaluation estate good enough to tell whether a change helped**, which is the binding constraint for almost everyone; and the balance sheet to carry the obsolescence risk when the frontier moves under you. Hold four and you cannot do it. That combination sits with the hyperscalers, the frontier labs, a handful of megacap technology firms, and a thin tail of banks, pharmaceutical companies and national champions with real machine-learning organizations. **And it is a continuum rather than a switch** — customer-owned weights and infrastructure, customer-owned data and orchestration on rented compute, managed dedicated training environments, private fine-tuning, vendor-operated but isolated customer state, customer-controlled retrieval and evaluation. Some of that is vertical integration; some is partial internalization; some is contractual governance of specialized infrastructure. **What they share is a direction: Nadella's answer moves the governance boundary inward, and the further inward it moves the more capability, cost and obsolescence risk you take back.** Verification is the other path — keep the specialization, and make the boundary checkable.

Now the strongest objection to what I just said, which Microsoft has already funded. If vertical integration is unaffordable for all but the largest firms, the obvious answer is to sell it as a service — and that is precisely what Frontier is. Six thousand experts going inside enterprises to build the loop turns a capability held by a few dozen organizations into one available to any company that can pay a consulting bill. **That is a serious counter and it is substantially right.** The forward-deployed engineering wave is a real investment in capability distribution, aimed at enterprises that could not have staffed such loops alone — whether the programs deliver that at scale is too early to establish. Nadella's prescription is being executed rather than merely written.

But look at who is holding the pieces afterward. The delivery organization that builds your learning loop works for the company that supplies your model. The infrastructure it runs on is theirs. The deployment knowledge accumulated in doing it — how your business actually works, which is the very asset the essay says you are paying with — is now also resident in their delivery organization. **You have bought help becoming independent from the party you were trying to become independent of.**

One honest asymmetry before I press the point. Engineers who learn your business inside an engagement agreement are covered by exactly the trade secret theory argued below — a named counterparty, a duty of confidence, a live fact pattern. The delivery organization is the *more* governable half of this arrangement, and I should not pretend otherwise. What it does not fix is the machine half, which is the one nobody can check.

With that said: this is not vertical integration. It is deeper specific investment wearing its clothes. Williamson's point was that integration solves holdup by putting the asset and the decision under one roof. Here the roof is the vendor's. The asset is more specific than it was, the switching cost is higher than it was, and the counterparty you would need to verify is the same one, now with a larger team inside your building.

And it still does not answer the question this article asks. A delivery organization can build you a boundary. It cannot give you a way to check the boundary held — not because its people are dishonest, but because a check performed by your supplier is the thing this whole article says is not independent verification. **The available rebuttal is separation of duties, and it deserves stating.** Nobody proposed the supplier check its own work: you buy the build from Frontier and the check from an audit firm, a rival integrator, or an attestation provider. I do not think that disposes of the objection — though the reason needs stating precisely, because the primitives an independent checker would need mostly exist. Attested execution is available today. What I have not found is a standard frontier-vendor assurance that binds an attested learning process to the identity and authorization state of the customer material it consumed, and that is the object an outside checker would actually need. **But the objection is against the absence of those mechanisms, not against Frontier, and I should not let the two merge.**

Frontier is a very large, very well-funded answer to a different question: how do I get this working? The question of how you would know what it did with what you gave it is untouched by six thousand engineers, and would be untouched by sixty thousand.

And it does not remove the trust problem, it relocates it. A firm running its own loop on somebody's rented accelerators, under somebody's model license, with somebody's inference endpoint in the path, has changed the shape of its dependence rather than ended it.

> **Vertical integration is what you do when you cannot verify. The interesting question is what it would take to verify — because verification, once built, is within reach of every firm, and integration is within reach of almost none.**

Read that as a friction claim and it sharpens. Owning the whole loop is a way of buying your way out of friction: you remove the counterparty you would otherwise have to check. It works. It is also the most expensive instrument on the shelf — which is exactly what a firm reaches for when the cheap instrument has not been built.

> **An answer only the largest firms can buy is not an answer to the problem. It is a description of who escapes it.**

Everyone else needs the cheap instrument. That is why a contract clause beats a data center for all but a handful of buyers, and it is what the rest of this article is organized around.

What the law already gives you

This is the section most discussions of this subject skip, and skipping it is why the problem looks larger than it is.

Trade secret law reaches use, not just copying. Under state trade secret law, and since 2016 the federal Defend Trade Secrets Act, misappropriation covers *use* of a secret acquired under a duty of confidence. You do not have to prove your document was reproduced, or that anything is extractable from a model. If a vendor took confidential operational know-how supplied under a confidentiality obligation and used it to build a capability, that presents a recognizable trade-secret fact pattern if the information qualified as a trade secret, reasonable secrecy measures were maintained, and the challenged use exceeded the recipient's authorization — and head-start injunctions and unjust-enrichment awards are established remedies in trade-secret practice, available long before anyone could inspect a neural network. The Federal Judicial Center's trade-secret case-management guide sets out the head-start approach and collects the cases applying it.

What it does not reach is the rest of the space, and the boundary matters: information that is generally known or readily ascertainable, an employee's general skill and knowledge, information the vendor independently developed, or anything otherwise failing the statutory elements.

And one reconciliation this section owes you, because the verification section below appears to contradict it. Relief from having to prove *copying* is not relief from having to prove *use* — and proving use directly, at per-customer granularity, is exactly the thing I argue below is unaffordable. What keeps the claim survivable is the ordinary circumstantial route that trade secret cases have always run on: proof of access, plus a capability appearing on the other side shortly afterward, plus the absence of a credible independent-development account. **That is a litigation posture, not an audit, and it is why the clause work matters more than the case law.**

And it does not reach a use you already licensed. This is the reconciliation the optimistic version of this section owes you. Consented use is not misappropriation. If your agreement contains a broad "aggregated or de-identified product improvement" carve-out — and one is present in every enterprise agreement in the sample I reviewed — then for that channel the vendor may have a strong contractual answer to a later claim that the use was improper. **Which makes the scope of the carve-out substantive rather than boilerplate** — whether the license actually covered this material, this use, and this purpose all remain live questions. **The carve-outs are not merely where a leak would be licensed. They may later give the vendor a contractual argument**

that the challenged use was authorized. Which is why the carve-out item sits near the top of the contract list below: the legal remedy is real, and its scope is set by a clause you can still negotiate.

Trade secret protection also requires reasonable measures, and here the exposure is internal. In *Trinidad v. OpenAI, Inc.*, No. 4:25-cv-06328-JST (N.D. Cal.), a pro se plaintiff's claim failed at the pleading stage for want of reasonable measures, on the theory that developing a protocol *by using a general consumer AI service* was voluntary disclosure. It is weak authority, it is not a final appellate holding, and it is distinguishable — the plaintiff had no identified applicable confidentiality restriction, whereas an enterprise arrangement may impose one depending on the actual agreement, the covered service, the data categories and the configuration — subject to the carve-out point above, which determines its scope. But the fact pattern that should worry you is the one it rhymes with: an employee pasting operational know-how into a consumer-tier tool outside your enterprise agreement, where no confidentiality obligation exists. Such a disclosure may weaken the organization's reasonable-measures showing for the disclosed information or related material, depending on the surrounding controls and the response.

Regulators have obtained model deletion without solving attribution. The FTC has secured algorithmic disgorgement — deletion of models or algorithms built on improperly obtained data — in *Cambridge Analytica* (2019), *Everalbum/Paravision* (2021), *Kurbo/WW* (2022), *Edmodo* (2023) and *Ring* (2023), on a provenance-and-consent theory that never required proving which parameter came from whom. **The orders are not all the same shape**, and the distinction matters: *Edmodo* and *Ring* expressly require deleting models or algorithms derived from the tainted data, while the contemporaneous *Amazon Alexa* order works the other way round — it requires deletion of covered data and bars its use to build or improve data products, rather than ordering an existing model destroyed.

State the posture precisely, because it is often overstated. These are consent orders and, in the Cambridge Analytica matter, an administrative order entered after the company failed to answer. None was litigated to judgment on contested merits, so these matters did not produce a contested merits determination concerning the legal basis, the permissible scope, the proportionality or the technical limits of model-level deletion on comparable facts. And each concerned alleged deceptive, unfair or statutorily noncompliant handling of consumer personal data — including unauthorized collection, use, access, disclosure or retention, or the failure to obtain legally required consent, with the children's-privacy matters brought under the COPPA Rule — rather than a paying enterprise's know-how supplied under a signed contract. So it does not establish what can be compelled. **What they do establish is that model-level deletion has been used as an enforcement remedy without first resolving parameter-level attribution.** They do not establish how far that remedy could be compelled against a contested respondent at frontier scale.

Its limit is who can use it. That specific form of FTC algorithmic disgorgement is a regulator's instrument, invoked rarely, and not something a private enterprise invokes for itself — depending on the claim and facts a private litigant may seek injunctive, deletion, contractual or other equitable relief, but these matters do not establish its availability with your vendor over a contract you both signed.

European data protection law gives you a purpose argument, and it is narrower than it is usually reported to be. Data collected to serve your request is not automatically available for model development: further processing requires a compatibility assessment or a separate basis. **The EDPB analyzes development and deployment as distinguishable stages** — but it says expressly that whether they constitute separate purposes and separate processing activities is a case-by-case determination on the actual facts, not a rule that they always are. **That same opinion accepted that legitimate interest can sometimes cover development**, so this is leverage rather than a bar — and leverage only where personal data is in the flow. If personal data is in the mix, you have more leverage than your vendor's sales team implies.

And for one channel there is a mathematical guarantee rather than a promise. Differential privacy with per-customer accounting bounds how far a trained model's behavior can differ depending on whether any single contributor's data was included. It requires no new right and no attribution, and it is the strongest formal guarantee discussed here **for bounding contribution-level parametric influence.**

Three qualifications, because this is the claim I am most tempted to overstate. It is a training-time mechanism, so it does nothing about retrieval over your corpus or persistent memory — the row I just said was least governed. If the question is how much one enterprise's participation can move the result, the privacy unit has to correspond to that enterprise rather than to an individual record, and the privacy budget that makes the guarantee meaningful can impose a material utility cost, mechanism and task dependent, which is one candidate explanation for its absence. And the guarantee is only as good as the unverified assertion that the mechanism was run, at the stated budget, with the stated unit — which is why it belongs *beside* attested execution rather than above it.

> **Friction, instance two, running the other direction.** Here the cost falls on the vendor rather than on you — accuracy, engineering, and a guarantee they then have to stand behind. I have not seen buyers routinely ask for it, and I have not found a standard frontier-vendor offering that incurs the cost. That is friction on the demand side, and it is one of four things in this article a reader can move without anyone's agreement — this, the planted canary below, the internal disclosure audit at the end of the clause

> list, and governing what leaves your own boundary. The cost of *asking* is one line in a procurement questionnaire.

So the honest position is that the gap is much smaller than the discourse suggests — and it is two gaps, not one.

> **The durable non-parametric layer has the least standardized legal and contractual treatment — existing doctrines may well reach a particular index, memory record or evaluation set depending on what it contains and how it was made, but I have not identified a general rule assigning the customer an exclusion right across the whole class of derived state. And across every layer, including that one, you cannot routinely, cheaply, and without litigation verify what your vendor actually did: a verification gap.**

I would rather lose the cleaner slogan than defend it under cross-examination. The second gap is the larger one, and it is the one in which I have not identified a standard frontier-vendor verification offering.

Why the existing vocabulary does not close it

Two standards efforts are usually offered as evidence that this is already handled. Both are real, both are more careful than their summaries, and neither has a word for the thing at issue.

The content-provenance label is richer than critics allow. The Creator Assertions Working Group's *Training and Data Mining* assertion (version 1.1, ratified May 2025) defines four entries — data mining, AI inference, generative training, and non-generative training — and separates the last two deliberately, precisely because generative training can produce new assets from the originals and object detection cannot. Anyone claiming it fails to distinguish generative training has not read §3.1.

Its actual limit is the object it attaches to. All four entries are labels a publisher signs onto **a file**. There is no entry for a feedback event, a trace, an eval suite, or a memory record — none of which is an asset with a manifest.

And the one place it could carry our distinction is a free-text field. Each entry is tri-valued — allowed, notAllowed, or constrained — and `constrained` carries a `constraint_info` string. The specification's prose calls the field `constraints_info`; its normative CDDL and its own example use `constraint_info`, and I use the normative name here. You could write "may be read to answer, may not be indexed" into it. **So the limit is not that the vocabulary cannot express the distinction. It is that expressing it lands in prose that no system parses, on an object that is not the one leaking.** That is a weaker criticism than the one usually made, and it is the accurate one.

The internet draft is now narrower than it was, and this matters for anyone citing it. Earlier revisions of `draft-ietf-aipref-vocab` carried an *AI Output* category alongside model production. **Revision -05 removed it.** As of **revision -07, published August 19, 2026**, the vocabulary defines exactly two categories: *AI Model Training* — "using an asset to modify the learned parameters of an AI model that is used to generate synthetic content in one or more modalities" — and *Search*.

Search is a retrieval category, and it is worth reading before assuming it covers you. It applies only where the application's primary purpose is to point users at assets, where results carry "a direct reference or link to the original location," and where excerpts serve to help a user judge relevance. It expressly "does not include the use of assets to generate summaries." An assistant answering a question over your corpus does none of those things.

So enterprise retrieval and persistent memory fall between the two categories rather than inside either. And the draft is careful about what that silence means: an uncovered use has no stated preference, and §5 adds that while one approach to an unknown outcome is to assign a default, "this document takes no position on what default might be assigned." **It neither permits the use nor prohibits it. It has no vocabulary for it, and declines to say what silence implies.**

And it disclaims enforcement in its own text. §3.2 states that the specification does not "ensure that preferences are followed," does not "address if, how, or when preferences should be followed or not-followed," and does not "address technical, legal, contractual, or other mechanisms that might create a stronger requirement to follow or not follow preferences."

Two cautions on citing it at all. The draft carries a notice on its face that its contents "DO NOT REFLECT CONSENSUS of the Working Group either in whole or part," and §3 and §4 each carry their own non-consensus note. It is a working document, not a settled standard, and quoting it as a standards-body position overstates it.

And the current draft text points the same way your contract already does — with the caveat above that this is one revision's text rather than a consensus position. §5.2 provides that "contractual agreements or other specific arrangements might override statements of preference." So even a category that did cover memory and retrieval would yield to the vendor agreement you signed. The override runs the wrong way for a customer whose contract already grants the use — which returns the entire question to the clause list at the end of this article.

Both instruments were also built for a different relationship: a publisher signaling to a crawler about a public asset. You are a customer inside a negotiated contract. Transplanting the vocabulary is not obviously wrong, but it is a transplant, and nobody has argued for it.

> **Friction, instance three.** A standard is a friction-reduction device: it makes a thing cheap to say and cheap to check, so parties stop negotiating it one at a time. Where a standard exists, the cost of expressing a preference falls to nearly zero. Here no standard covers the channel, so every enterprise that wants this must draft it, argue it, and pay for it alone — which is precisely the condition that keeps a practice rare and expensive.

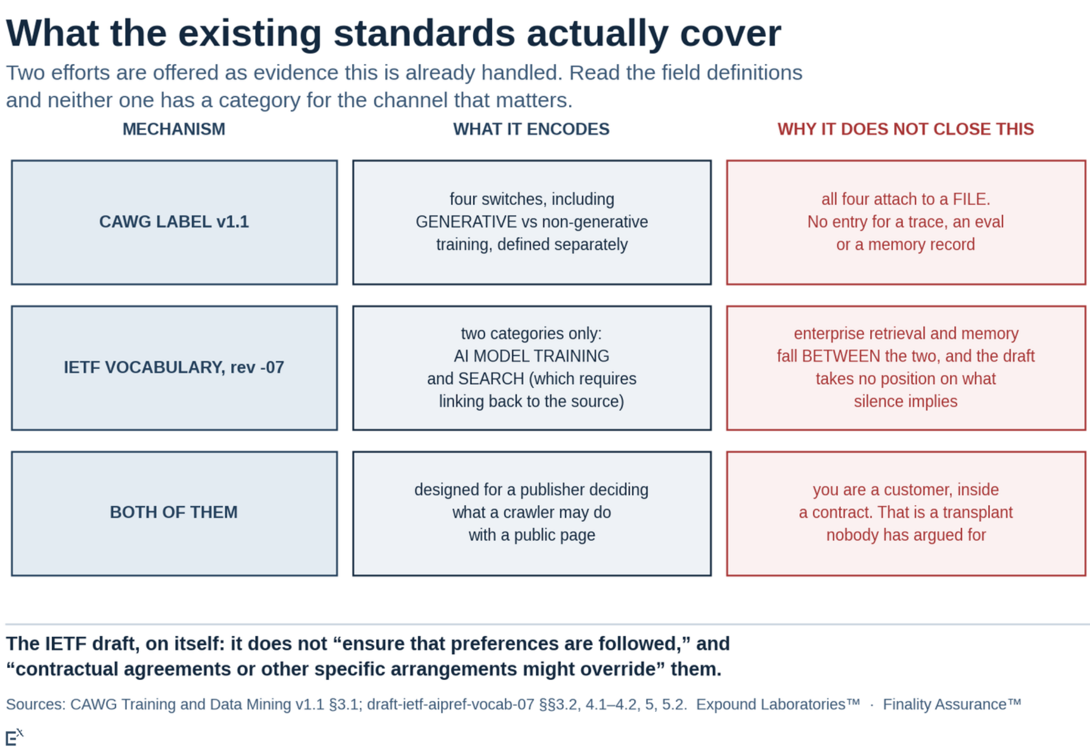

Figure 4 *What the existing standards actually cover*

Three answers arrived, and each guards a different door

Elacity Labs proposes encryption with terms written into the package and keys released only when rights check out, split across a group so no single party can quietly release them. Their criticism of the tenant approach is correct — *"a promise from your landlord, enforced by policy and revocable by policy."* **What it guards: opening the file.** It does not address a model answering over your data. **But it has an architectural property nothing else here has, which matters later.**

NeoSapients maps its platform onto Nadella's five controls: *"You should be able to run the best models in the world without teaching them the things that make you, you."* **What it guards: where it runs** — real, and current best practice, and a restatement of existing procurement rather than a new instrument.

Marisetty and Mamidi formalize the problem and conclude that *"a single corrective instrument is a subsidy under under-retention and a tax under over-retention."* **What it guards: nothing — it prices.** That is a legitimate answer and a different instrument for the same failure. Whether pricing substitutes for a right depends on transaction costs being low and the entitlement being well defined; this article argues the second condition currently fails, which is a reason to fix the entitlement rather than a reason their result is wrong.

The verification problem, stated honestly

The instinct is to check afterwards. Did my knowledge end up in your model?

This is harder than it looks in general, and easier in one specific case. There is an active research field devoted to exactly this: several distinct methods for asking whether a particular example influenced a particular model. None of it is vaporware.

The cheapest case that can produce useful positive evidence is the planted canary. Seed a distinctive, meaningless string through your material, then check whether a later model reproduces it. That is cheap, requires no special access, and produces a result a non-expert can evaluate.

Its sensitivity depends on how the canary is inserted, on the model and training regime, and on the preprocessing and deduplication pipeline — some deduplication systems will suppress a repeated pattern and others will not, and the answer differs between bulk pretraining corpora and the instruction and preference sets where customer material would more plausibly land. **A negative result remains weak**: it is equally consistent with your material never being used, with it being used and not memorized, with the canary being filtered, and with an insensitive test. **A positive result can be informative.** That asymmetry makes it a lottery ticket rather than an audit — worth buying because the ticket is nearly free.

Two practical conditions. Use matched control strings, seeded at the same rate and never exposed to the vendor, or a hit tells you less than it appears to. And seed into synthetic records rather than live ones, and check the vendor's acceptable use terms first: a canary planted in real material creates a processing purpose your own privacy notice probably does not describe, and a canary that does get memorized is extractable by third parties.

Everything past that gets hard fast, and none of it gives you a contractual remedy.

> **Friction, instance four — and this one is mixed, which the taxonomy has to be allowed to say.** Notice the form of the objection: not *this cannot be known*, but *knowing it is noisy, expensive, contestable, and requires access you will not be given*. Every term in that sentence is a cost. None is a prohibition. One reason attribution is unattractive as the primary proof theory is not that the evidence is impossible — it is that the evidence costs more than the claim is worth, which is the definition of a right made safe to ignore. **Attribution at per-customer granularity is noisy, expensive, contestable by any competent expert, and generally requires access to the model and the training pipeline that your vendor will not give you. And attribution is not one category — it is method-, granularity- and proposition-dependent.** In instrumented, smaller or specially structured settings the method exists and access, expense and contestability dominate: that is friction. **At the instance level against large-model pretraining**, published evaluations have found membership-inference attacks performing at or near chance across most tested settings, with apparent successes explained by distribution shift between the member and non-member sets rather than by a membership signal — *Duan et al., Do Membership Inference Attacks Work on Large Language Models?*. **There the limitation is methodological rather than merely economic**, which by this article's own test is nearer a technical gap than a price.

But that is a claim about instance-level membership, and it does not generalize to all attribution. Dataset-level methods aggregate evidence differently and can separate training corpora where per-record tests cannot — *Maini, Jia, Papernot & Dziedzic, LLM Dataset Inference: Did you train on my dataset?* — and newer black-box corpus-level work reports positive results. **So the procurement question is not whether a literature exists**, and not whether "attribution works" in the abstract. It is whether an available method, at the granularity your proposition actually requires, can answer it with enough accuracy and resistance to expert contest to support the reliance you intend. A taxonomy that cannot return an answer inconvenient to its author's thesis is not a taxonomy. Good enough for research; generally too noisy and contestable to carry a claim by itself — though it can contribute alongside access, timing, similarity, internal documents and discovery, which is how circumstantial cases are actually built.

So the useful move is to shift the check earlier — to the point where an experience is offered as an input, before anything changes.

And here the honest version is much weaker than the clean version, so here is the clean version and then the correction.

The clean version says: at the moment of admission, everything you need still exists — what the experience is, where it came from, what change is proposed. **Most of that is false.** Under ordinary training there is no per-example "proposed change"; batches are shuffled, examples are deduplicated and rewritten, provenance is routinely lost, and a trace can be laundered into synthetic training data that carries no marking at all. The admission boundary is not a clean gate. In continuous and online systems it is not even a well-defined moment.

What survives is lot-level, not example-level. A vendor can commit, before a training run, to which corpora entered it, with a manifest that names the excluded sets. That is coarse. It is dramatically better than nothing. **The mechanism is available today for bounded corpus identities**; its engineering cost and operational feasibility at frontier scale, and for training material generated during a run rather than assembled before it, remain empirical questions.

> **The inability to unscramble the omelet is not an argument against recording which eggs went in.** Though it does matter a great deal who writes the list of eggs — which is the subject of a later section, and the reason a manifest is a starting point rather than an answer.

And a corpus commitment governs a narrower object than the problem does. It applies most directly to bounded, batch-like learning events: a run with a corpus assembled beforehand. It does not, by itself, govern serving-path memory and retrieval, human review, system-prompt construction, continuously updated routing, or preference and reinforcement stages where much of the training material is generated by the model *during* the run and therefore cannot be committed to in advance even in principle.

Which points at a second governable object. For those channels the thing you can fix in advance is not a final corpus but a **rule**: which sources are admissible, what may be retained, what it may be used for, what may be collected at all. Corpus admission and ongoing state-and-use admission are different instruments, and the honest position is that the first is the one that exists today — while the routes this article spends most of its length on are governed by the second.

That gap is worth stating plainly, because it cuts against the tidy version of my own argument. The middle row is where the governance is thinnest, and the technical instrument available today reaches most cleanly into the row that is already argued about. The middle row is currently governed by contract, which is why the clause list at the end of this article is the practical deliverable and the manifest is not.

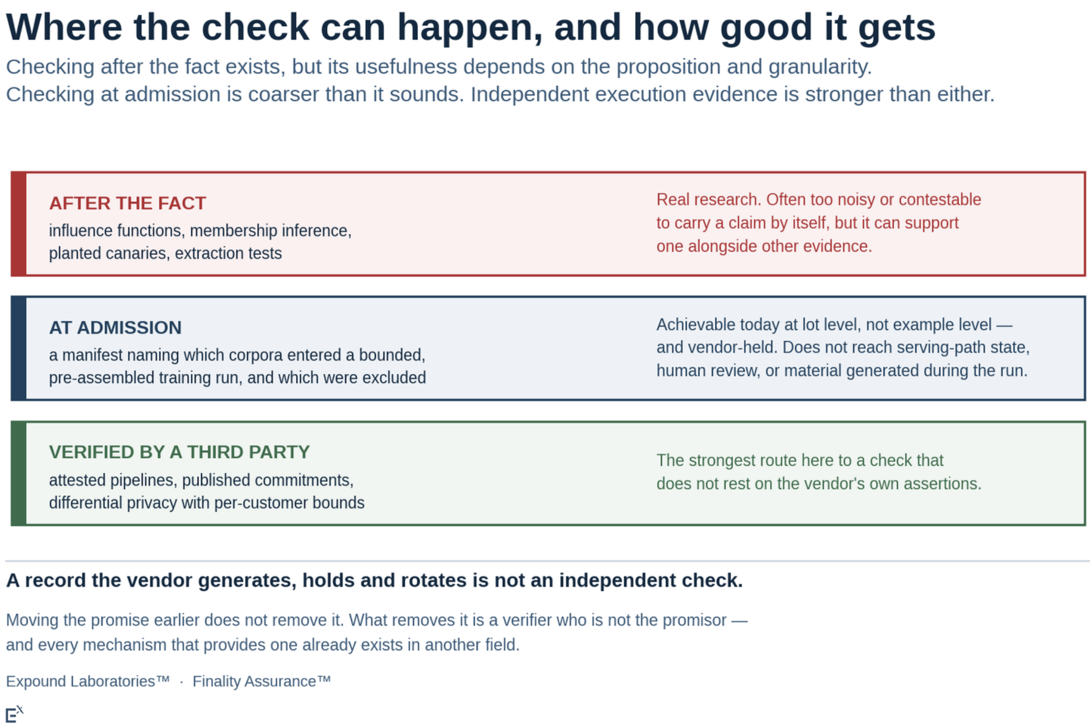

Figure 5 *Where the check can happen, and how good it gets*

What this costs either way

There is a ledger on both sides, and the side that cuts against me could be the larger one — which I cannot establish, for exactly the reason I cannot establish the claim it argues against. Leaving the gap open leaves the contributing firm unable to price its own exposure. Closing it costs everyone else.

Arrow, in 1962, on why fixing knowledge markets also hurts: *"precisely to the extent that it is successful, there is an underutilization of the information."*

That cuts against me and I will state the cost plainly. **A learning-permission regime makes AI systems worse for everyone who is not party to it** — to whatever extent corrections currently reach shared models at all, which is precisely what I have said nobody can establish. Fewer corrections would propagate. Failure modes found by one customer would go unfixed for others. Improvements that had been free would require a negotiation. **Note the bind, because it is symmetric and I do not get to escape it:** this cost is large only if the hypothesis is true, and the same opacity that blocks the claim blocks the objection to it. Anyone proposing this owes an account of that cost, and the account is: the loss is worth bearing only if the alternative is that the good contributions stop entirely.

Which is the argument, and it is a prediction rather than a finding. If every expensive correction improves a vendor's asset and reaches your competitor, the rational response is to stop making expensive corrections. Withhold feedback. Keep your tests internal. Move the valuable work somewhere sealed.

I want to be careful here, because the evidence currently runs the other way. Enterprise AI adoption is rising, not falling, and no published data shows contribution withdrawal at scale. This is a mechanism I expect, not one I have measured. If by 2028 no enterprise has publicly restricted feedback, evaluation-set sharing or telemetry on this rationale, and none of the buyers I work with has done so privately, I am wrong about the size of the effect.

And the objection that lands closest to home, which I have not seen made: if corrections travel, they travel both ways. The same mechanism that would carry yours to a competitor would carry every other customer's to you. Which direction is worth more to you is unestablished — nobody outside a vendor can price either side, which is the whole difficulty. **The firms with a real grievance are the smaller number whose accumulated operational judgment is a competitive asset rather than an ordinary cost of doing business.** If you are not one of them this is still worth an hour: the clause work costs a meeting and the internal audit costs nothing. **Knowing which firm you are takes an afternoon, and it is an afternoon I have rarely seen budgeted.**

And the economics of why now is genuinely unresolved. Harold Demsetz argued in 1967 that rights emerge when technology makes an externality worth internalizing and cheap enough to internalize. AI has arguably done both, and the right is not emerging.

That is not yet evidence against him. Demsetz gives a functional account with no mechanism and no clock — the standard objection to the paper, and the reason it is often called the naive theory of property rights. His own case study runs over decades. Two years of absence falsifies nothing.

It does make the mechanism the interesting question. My answer is coordination failure: no vendor gains by moving first, and no customer can demand it alone. That is an assertion I can motivate and have not demonstrated, and it is the part of this argument I would most like to be shown wrong about.

The limit on any answer

Michael Heller described the tragedy of the anticommons in 1998: when too many parties each hold a right to exclude, the resource ends up underused, because every one of them can say no and only unanimity produces a yes.

The transplant is imperfect and worth naming. Heller's cases are rival resources — Moscow storefronts with too many licensing authorities. A training corpus is non-rival; my use does not consume yours. What survives the difference is the veto structure: a training run is a joint production process that requires assembling many inputs at once, so a holdout blocks the run rather than the resource.

And it applies to any scheme of this kind, mine included. Reconciling a lot-level manifest against thousands of individually negotiated exclusion scopes would produce exactly the failure Heller describes — a training run gated by its most restrictive customer. **Which is why the exercise has to cost machine time rather than legal time**, and why declaring obligations in a machine-evaluable form, rather than in prose, is load-bearing rather than decorative.

The current draft text reaches for the same conflict rule, in a narrower setting. `draft-ietf-aipref-vocab` §5.1 resolves conflicting statements about a *single asset from different declaring parties* so that "the most restrictive preference applies." That is not a rule for reconciling thousands of customers across a corpus, and I do not want to claim more from it than it says. What it shows is that the restrictive-wins rule is the natural one, and the restrictive-wins rule is the anticommons written as an algorithm. Which means:

> **This works only if exercising the right costs machine time rather than legal time** — standard classes, sane defaults, no bespoke negotiation.

One clarification I owe on the frame. I have been calling this a permission rather than a property right, on the grounds that it does not claim ownership of knowledge. That distinction is thinner than it sounds, but it is not the distinction between property and not-property: **a veto over how a lawfully held input may be used is exclusionary in the economically relevant sense, and that does not require treating the knowledge, the model or the derived state as anybody's property.** I defend it on the ground that its scope is narrow — it reaches the experience you contributed under your agreement with that vendor, and nothing else.

So what is actually missing?

Not ownership of knowledge — that loses every argument about facts and independent discovery, and deserves to. Not ownership of the improved model; the provider built that.

What is missing is narrower than the right being asked for, and harder than a promise:

> **A way for a customer to verify, without trusting the vendor and without litigation, that their contributed experience did not participate in a change they did not authorize.**

Three mechanism classes address different parts of this, and conflating them is a mistake the rest of this article has been careful to avoid. Admission controls can support a *non-entry* proposition. Attested execution can support claims about *what code ran*, and does not by itself establish corpus identity. Differential privacy assumes participation was permitted and bounds *influence*. Taken together: **auditable controls over admission — evidence that your material never entered a shared learning process — attested execution of the pipeline, and per-customer differential privacy where you do authorize learning.** The first is the one to ask for, because it is the narrowest object that answers your actual question. **You do not need the vendor to disclose everything that entered its corpus. You need evidence that yours did not** — a far more precise ask, and one a vendor can meet without exposing third-party licensing, security or competitive information. The qualification on why the third is not sufficient alone is in *What the law already gives you*, above. **Individually, these are not exotic and some are shipping products — attested execution most clearly. What I have not identified is a standard frontier-model-provider enterprise offering that combines them into customer-verifiable evidence that a particular customer's contributed experience did not enter an unauthorized shared-learning process** — nor have I seen that assurance asked for in the procurement language I have read, or used as a differentiator. **That is an observation rather than a survey, and one counterexample would correct it.** That is a market-structure and engineering vacancy — the one a buyer can act on without waiting for anybody.

Eleven issues to raise with counsel in your next AI vendor agreement

One reframing first, and it is worth more than the rest of this section combined.

Do not negotiate a *no training* covenant. Negotiate a **no shared learning** covenant.

> **"Training" is too narrow an object to carry the weight buyers put on it.** Your material can change a shared service through fine-tuning, distillation, evaluation and model selection, safety-classifier development, routing policy, retrieval corpora, exemplar construction, synthetic-data generation or preference learning — none of which needs to meet somebody's definition of *training*.

A vendor can promise not to train the foundation model, and truthfully keep that promise, while your conversations improve a router, seed an evaluation set, tune a safety classifier and select a better post-training recipe. **From your side of the boundary that distinction is invisible and, mostly, irrelevant.** So govern the *effect*: whether information you supplied for your own service can become learning that alters a shared system for somebody else's benefit.

That reframing is also the one this article has been arguing for throughout. The middle layer — memory, indexes, evals, routing — is where governance is thinnest. A clause list organized around the word *training* governs the layer everyone already argues about and leaves the thin one alone.

Before the list, three points of precision.

Be precise about who. Everything below belongs in the agreement between your company and your AI vendor, in the wide sense defined at the top of this article.

Find out which document the language lives in. These terms scatter across a master services agreement (the MSA), a data processing agreement, an acceptable use policy, and product terms the MSA incorporates by reference. **A no-training assurance in a webpage the vendor revises unilaterally is not the same instrument as a negotiated clause in a signed agreement.** Ask which yours is before you rely on it.

And do this per supplier, per service, per feature, and per data path. One vendor can treat API inference, enterprise chat, stored conversations, files, retrieval indexes, prompt caches, fine-tuning, feedback, grounding and zero-retention configurations differently — and the same model bought through a hyperscaler can travel a different path from the same model bought direct. **Follow the information, not the logo. For each path your sensitive material actually travels, your**

exposure is bounded by the least protective applicable agreement, configuration and downstream processor on that path.

> **This is not hypothetical, and it is checkable today.** Google's published Vertex AI documentation commits that customer data will not be used to train or fine-tune its models without permission or instruction — and, on the same page, sets out feature-by-feature retention that varies inside the one product: prompt logging for abuse monitoring, thirty-day storage for Grounding with Google Search and for Grounding with Google Maps with **"no way to disable"** it, twenty-four-hour session caching for the Live API, and in-memory caching of inputs, outputs **and derived data** that is on by default and must be switched off per project. One vendor, one product, five different answers. **That documentation establishes the terms and technical representations Google publicly makes about those features — which is a document you hold, and clears the bar this article set.** It does not independently establish that the infrastructure behaved as represented in any given instance, and I am not going to switch standards because the switch would be convenient here.

Eleven issues, in the order I would raise them. These are negotiation objectives rather than model clauses — governing law, bargaining power, service architecture and your own data will all change the drafting, and this is a list to take to counsel rather than to a vendor. Several will get a no; I have noted the fallback where the no is likely, because a request with no fallback is a wish.

> **One thing this list is not.** These eleven issues are an **overlay** on the statutory, regulatory and sector-specific terms your relationship already requires. They are not a substitute for a compliant data processing agreement, a business associate agreement where health information is involved, a lawful international transfer mechanism, security and breach-notification terms, allocation of AI-specific regulatory roles, or any other mandatory provision applicable to your organization. **This list is about a gap those instruments do not address.** It assumes the rest is present, and it is not a compliance checklist. **None of it is legal advice**, and the drafting belongs with counsel who knows your jurisdiction, your sector and your data.

1. **Define the protected material in three disjoint classes, plus one transformation rule.** *Deidentification is a condition information can be in after processing, not a fourth class of provenance — treating it as a category is how it becomes an exit.* Most agreements protect "Customer Content," meaning inputs and outputs. Split it. **Protected Customer Data**: inputs,

outputs, uploaded files, fine-tuning source data, feedback and evaluations you supply, traces to the extent they contain your content, and metadata that itself reveals it. **Operational Service Data**: token counts, latency, error rates, billing — legitimate to use, subject to confidentiality and minimization. **Customer-Derived Data**: what the vendor generates *from* that material — embeddings, vector and retrieval indexes, persistent memory, conversation and agent state, machine-generated annotations, and customer-specific adaptive state. **Say expressly that the restrictions here concern use, disclosure, persistence and shared learning, and do not by themselves determine ownership of the technology that generated or stores them.** *You should not have to win an ownership fight over an embedding in order to restrict what is done with it* — and you do not need to own the improved system to restrict certain uses of what you contributed.

Then solve the problem every three-class taxonomy actually has, which is that real artifacts are composites. A single trace can carry your content, a machine-generated annotation and a latency figure at once, so an instruction to keep the classes disjoint is unsatisfiable for the artifact that matters most. **Classify separable components independently where that is practicable. Where an artifact contains material from more than one class and the components cannot be reliably separated, the most protective applicable treatment governs. And anything not expressly classified defaults to Protected Customer Data.** That residual rule is the important one: without it, every ambiguous artifact is resolved by whoever argues harder, at the moment it matters.

Define Operational Service Data narrowly, and preferably as a closed list. It is the class marked *legitimate to use*, which makes it the exit. Do not let feedback signals, accept and reject decisions, semantic state, customer-specific annotations or customer-derived embeddings become Operational Service Data because somebody can plausibly call them telemetry — your accept and reject signals are, on the list earlier in this article, one of the routes by which durable state crosses the boundary.

And the transformation rule, which is not a fourth class: treat deidentification or aggregation as a **permitted transformation, not an automatic exit from protection.** Broader use should depend on agreed objective criteria — no party, using means reasonably likely to be used, can identify the customer or an individual, reconstruct the protected material, extract substantially similar customer-specific confidential information, or re-associate it with you — together with a defined assessment process and an express allocation of who bears the burden of demonstrating those criteria are met. **The test should not be what the transforming party can satisfy itself of.** And say expressly that contractual criteria do not displace statutory definitions or the deidentification standards applicable to regulated information; where those apply, they are the operative test and this clause sits on top of them.

1. **Prohibit shared learning absent enterprise-level authorization.** Define a **Shared Learning Process** as any process by which your protected or derived material is used to train, retrain, fine-tune, post-train, distill, reinforce, evaluate for the purpose of improving a model or system, select training or evaluation data, generate synthetic training material, construct exemplars, alter routing or ranking policy, enrich retrieval corpora, improve safety or moderation models, or otherwise modify shared parameters, policies or adaptive state. Then prohibit it without your express prior written authorization. **This is the clause that survives a change of terminology. And define the other side of it, or you will forbid what you are buying:** a **Customer-Specific Learning Process** is one whose resulting state is technically and contractually isolated to you and is not used outside your agreed environment and does not modify a generally available service. *"Benefits other customers" is the intuition; "used outside your environment or to modify a generally available service" is the observable test, and it is the one to put in the contract.* Your own fine-tune, adapter, memory, index or routing policy all *train* something. **The objection is not to learning. It is to learning that creates a reusable benefit outside your boundary.**
2. **Bar individual users from granting that authorization.** No thumbs-up, thumbs-down, bug report or support ticket by any employee constitutes your company's consent to use protected material for model development — authorization comes only from a named administrator, in writing or through a designated control. **This is administrable, not aspirational:** Anthropic's published documentation says commercial inputs and outputs are not used for training by default but that explicitly submitted feedback may be, with the related conversation retained for up to five years — and gives Team and Enterprise owners a setting to disable feedback submission outright. *If your vendor cannot tell you which control does this, that is your answer.*
3. **Get a written data lifecycle schedule, feature by feature.** For each feature: what is created, why, where it lives, who can access it, whether it is customer-specific, whether it is ever combined across customers, whether it is used for improvement, default retention, whether you can delete it, backup and disaster-recovery deletion period, legal-hold and regulatory retention exceptions, your right to return or export before deletion, treatment on termination, and whether certification is available. **Name the exceptions or "delete on termination" quietly becomes "deleted from active systems, except backups, security logs, legal holds and support systems." Retention and learning are different questions and you need both** — a vendor can delete your input and keep what it changed, or retain material it is contractually barred from ever learning from.
4. **Separate permitted operations from prohibited secondary use, and enumerate both.** Permitted: delivering the service, fraud and abuse detection, security, legal compliance, billing, support you asked for — **as reasonably necessary to that interaction, your account,**

and the integrity of the service. Draw the line explicitly, because this is where vendor counsel will push: permitted operational security processing does *not* include using your protected material to train or improve a shared security, safety or moderation model, unless you separately authorize it for a narrowly defined abuse-prevention purpose. **Operational safety processing and shared safety-model improvement are different things, and a vendor will otherwise read the first as licensing the second.** Prohibited: everything in the Shared Learning Process definition. **Stop calling the second list "carve-outs from no training"** — that framing concedes that the vendor's definition of training is the operative boundary. *If refused:* narrow the product-improvement exception and time-limit it — **but do not settle for the words "security and reliability" alone.** *Reliability* can be read to cover hallucination reduction, routing optimization, quality improvement and preference learning, which rebuilds most of the right you just removed. Bound the fallback by **purpose, data, retention, resulting state and recipients.**

5. **Flow every substantive restriction downstream, not just the security terms.** To affiliates, subprocessors, infrastructure providers, model providers, human-review and annotation vendors — anyone who receives the material, on terms no less protective *as to use*, not merely as to security. Where personal data is involved, processor arrangements commonly already require confidentiality, security and subprocessor obligations — **and none of that carries your no-shared-learning restriction**, so a generic flow-down tells you nothing about the thing you actually negotiated. Ask for the subprocessor list, advance notice of additions, an objection period on genuine privacy, security or regulatory grounds, and termination of the affected service without penalty if an objection cannot be resolved. **And make the vendor responsible for the acts and omissions of every downstream recipient as if they were its own** — otherwise the answer to a breach is *we required them to comply; take it up with them*, and you are chasing your vendor's subcontractor. **An absolute veto over every new subprocessor is generally not commercially realistic with a hyperscale provider; this structure is.**
6. **Fix precedence and prohibit unilateral expansion of data-use rights.** The negotiated data-use terms control over online terms, service terms, product documentation and anything published later. **Do not try to freeze all incorporated documentation** — providers genuinely need to change security procedures, acceptable-use controls and technical descriptions, and asking for a total freeze invites a no. The narrow ask is the one that matters: **no unilateral change may materially expand the vendor's rights to use your protected material, or materially reduce your agreed isolation, retention, deletion, objection or verification protections.** A vendor can leave its nominal rights untouched while extending retention, weakening isolation or changing a default — close both directions. Add the architecture case as well as the document case: **a materially new processing purpose, persistence mechanism, recipient class or shared learning process requires advance notice, and does not become**

permitted merely because you started using a new feature. A meticulously negotiated clause is worth little if an incorporated web page can change the data flow underneath it. Google's own Vertex AI documentation now carries a banner saying it is no longer being updated and has moved to a differently-named platform — which is exactly the mechanism this item exists to survive.

7. **Name the consequence, and match the liability to it.** Certified deletion of derived artifacts and termination for cause are reasonable asks. **Rolling back or retraining a shared frontier checkpoint generally is not** — it may be disproportionate or technically infeasible, and asking for it as the default remedy invites a no that costs you the rest of the clause. For a customer-specific fine-tune it is realistic. Ask instead for a remediation sequence: cease, preserve, investigate, identify affected datasets and model versions, quarantine and delete affected source and customer-specific derivatives, exclude from future training, apply unlearning or retraining where feasible and proportionate, certify in writing, and provide termination and transition rights. **Then spend your capital on the liability regime** — a supercap or cap exclusion for **first-party** losses from prohibited data use and confidentiality, privacy and security breaches, together with **indemnification for defined third-party claims**, which is where indemnity actually does its work. Add equitable relief for prohibited-use and confidentiality breaches, subject to law and facts. *A restriction whose breach falls into an ordinary low liability cap is economically much weaker than its wording suggests. Negotiate the consequence alongside the promise.*
8. **Ask for verification, and be precise about which kind.** A SOC 2 or ISO certificate is a representation with an auditor attached. **Ask which control objective was actually tested**, and whether any of them scope the learning pipeline rather than the corporate perimeter. Then ask the question that matters: **does the vendor maintain controls reasonably designed to prevent your material from being admitted to a Shared Learning Process without authorization, and evidence sufficient to determine afterwards whether it was?** *That is the contractual fact you care about* — evidence of **non-entry**, a far more precise and obtainable object than disclosure of everything else that entered. **Say plainly that the control covenant supplements the prohibition rather than qualifying it.** Otherwise a vendor can later argue it complied because its controls were *reasonably designed*, while your material went into the pipeline anyway. **"Reasonably designed controls" must not become a safe harbor for actual unauthorized use.** State them as two obligations that stand or fall independently: **the outcome covenant** — the vendor shall not perform unauthorized shared learning — and **the control covenant** — the vendor shall maintain specified controls reasonably designed to prevent and detect it. Otherwise a vendor argues either that good controls excuse the use, or that no proven use excuses the weak controls.

9. **Require the configuration to match the contract.** The negotiated restrictions must apply **by default across your organization**, and no ordinary end user may disable or override them. Any feature requiring broader processing must say so before activation and require an authorized administrator's affirmative act. **A negotiated right that an employee can switch off by enabling a feature is not a control.**
10. **If personal data is in the flow, contract the purpose limitation rather than arguing it.** Where the vendor is your processor, it processes only on your documented instructions and only to provide and secure the service — not for its own model development or product improvement without your written authorization. **And do not let a label settle the question.** Where the vendor determines its own purpose, or the essential means, of processing for model development, it may be acting as an independent or joint controller for that processing, depending on the facts and applicable law. Where it does, require that processing to be identified expressly — the activity, the purpose, the data categories, the legal basis, the retention and the recipients — before it begins. **Contractual labels do not determine the legal characterization of the processing**, and the direction that matters runs the other way: a vendor that uses your material for its own model development is likely making itself a controller as to that processing by conduct, whatever the agreement calls it. Which is also why the further-processing point earlier in this article is the sharper instrument here — using material collected to serve your request for a different purpose needs its own basis, and a disclosure that the vendor is doing so may be closer to an admission than a resolution. **Unlike a negotiated preference, purpose limitation can carry independent statutory consequences where data protection law applies** — and a negotiated term may also provide a direct contractual remedy between the parties without waiting for supervisory enforcement, subject to the governing law and the agreement's enforcement provisions.

And where you deliberately authorize learning, negotiate the boundary itself: which material, which system receives it, for what purpose, at what privacy unit, under what retention, whether the resulting state is customer-specific or shared, and what happens when the authorization ends. **If differential privacy is offered, ask for ε, δ, the privacy unit, the accounting and composition method, and the cumulative or lifetime budget — rather than accepting the label.** Every one of those does work. Record-level, user-level and organization-level guarantees are radically different things. ε alone tells you almost nothing without δ: in approximate differential privacy δ is an additive slack term on the ε bound, so it has to be read together with the mechanism and the accounting rather than treated as a standalone failure rate. And privacy loss composes across repeated mechanisms, so a per-run figure without a cumulative budget is an accounting statement rather than a guarantee. Note what it does and does not do: it can bound how much the output distribution of the trained mechanism changes depending on whether the defined privacy unit participated, under stated assumptions. **It does not establish that your material was**

never used, and if what you need is *no learning at all*, applying differential privacy to learning from your data solves the wrong problem.

And there is a tension worth naming, because it cuts against the framing of this whole article. If the privacy unit is your organization, then a guarantee tight enough to assure you that your corrections did not move the model is, by the same arithmetic, tight enough to assure you they did not help anyone — including you. The mechanism that would prove your contribution did not travel is in some tension with the premise that your contribution was worth something. That is not a reason to reject it. It is a reason to know which of the two you are buying.

And what to do when the answer is no, which for items 7 and 9 it often will be. **With one distinction worth keeping straight**: some of what sits near those items is a discretionary ask you can trade away, and some of it is a statutory floor your own regulator expects you to have secured. Audit and inspection rights over a processor are the usual example. Expecting a refusal is realistic; accepting one on a term the law requires you to obtain is a different problem, and it is yours rather than the vendor's.

Do not treat a refusal as the end of the transaction. Treat it as the output.

Get it in writing, dated, and specific. Naming who declined and when. **Be precise about what that establishes:** a written refusal does not establish what happened inside the pipeline. It establishes exactly which assurance the vendor was unwilling or unable to give you — which is a fact about the negotiation rather than about the system, and it belongs in your risk record on that basis.

Give the refusal a price in your scoring. If verification appears nowhere in procurement scoring, a supplier is rational to keep refusing, and no individual buyer changes that by asking once.

And aggregate it, because the diagnosis implies the remedy. If nothing is offered because no vendor gains by moving first and no customer can demand it alone, the instrument is not a better clause. It is several buyers asking the same question in the same words in the same quarter. Industry groups have moved supplier practice on security questionnaires, model cards and subprocessor disclosure by exactly this route. **The coordination failure I named is the kind a purchasing consortium is built to break.**

And one internal item, which needs nobody's agreement: decide which of your interactions you would mind a competitor benefiting from, and find where employees are pasting operational know-how into consumer-tier tools outside your enterprise agreement. Nothing in your stack currently distinguishes high-value interactions from the rest — and the disclosure that would cost you the legal remedy may not run through your vendor at all.

The objection everyone else leaves standing

You now have the clauses, and they are worth having: a negotiated scope is enforceable, and in the agreements I have reviewed buyers routinely leave that protection unexpressed. **What no clause on that list delivers on its own is proof.** Here is the objection, because it is the one that decides whether any of this is real.

A record of what entered a training run sounds like the difference between an assurance and evidence. **Held the ordinary way, it is not.** If the record is generated by the vendor, written by the vendor, formatted by the vendor and rotated by the vendor, it is the same kind of object as the clause it was meant to improve on: a statement about an internal process you cannot observe. **A log file generated, held and controlled exclusively by the vendor may well be evidence. It does not become an independent check because it carries timestamps.**

And this is precisely where the industry's current answer runs out. Microsoft has committed $2.5 billion and 6,000 experts to going inside enterprises and building the loop; AWS, OpenAI and Anthropic funded equivalents within weeks. That is an enormous, well-aimed effort at *how do I get this working* — and, as the Frontier section argued, it cannot touch how you would know what was done with what you gave it. A check performed by the party that built the thing may be perfectly good evidence. What it is not is independent verification of the proposition that party controls.

So the requirement is structural, and it is stateable in one line. The record has to be produced by a process the holder cannot quietly write, and it has to be reconstructable by somebody who does not trust the holder. **That is a design constraint, not an aspiration, and it is the constraint the machinery in the next section was built to satisfy.**

And the constraint splits cleanly into two halves, which is the practical point of this whole article.

The half you control, you can have today. Governing what leaves your boundary — which material is admissible for which purpose, under whose authority, with what evidence, and what you will accept back as relied-upon — is a decision on your side of the line. Run under a computed verdict, it produces a reconstructable record of **what you exposed, what you authorized, what evidence supported that decision, and what you relied upon.** That does not prove what your vendor did with it — that proposition sits inside the vendor's infrastructure. **It establishes precisely what was available to be done**, which is the predicate for every claim, negotiation and audit that follows, and no supplier has to agree to any of it.

It is also the answer to a problem you already have and may not have priced. Trade-secret protection turns on reasonable measures, and the fact pattern in the case discussed earlier — material developed through a service, outside the agreement anyone thought was governing — is a reasonable-measures failure before it is anything else. A disciplined record of what left your boundary, under whose authority, on what evidence, is directly responsive to that. **Your existing egress tooling was built for a different question**: data-loss prevention asks whether a transfer should be blocked, not whether a disclosure was authorized by someone competent to authorize it, on evidence that survives being read by a stranger later.

The other half is the vendor's to run. Admission controls over its own learning pipelines, attested execution, a per-customer privacy budget — each is something the vendor does *to its own infrastructure*. No amount of engineering on the customer side substitutes, because the object being evidenced sits inside the vendor. **Which is precisely why it is a market and procurement vacancy containing a bounded assurance residue.**

And the strongest objection to my framing lives right here, so let me put it rather than wait for it. Is attesting a frontier-scale training run a matter of price at all?

Less of a barrier than I first assumed, and I should say so against my own interest in a tidy story. Confidential-computing modes and remote attestation for modern accelerator infrastructure are shipping products rather than research proposals, and attestation composes: you collect a measurement per node and bind the measurements into a manifest for the run. What remains genuinely empirical is the engineering *cost and scope* of doing it across a frontier training fleet — the performance tax, the size of the trusted computing base once you include the scheduler, the orchestrator and the storage path, and the provenance binding discussed next. Those are engineering, cost and scoping problems. They are not an unknown scientific primitive, which by the test set out at the start of this article puts them in the friction column and not the gap column.

The residue is narrower than "binding" makes it sound, and worth stating precisely. Attestation establishes properties of the *software and hardware that executed*. It does not by itself establish that the program was correct, that it implements the policy you intended, or **which corpus it was handed** — and if the vendor authored the binary, an attestation proves the vendor ran the vendor's own code. **Cryptographically binding a declared manifest to an attested execution is engineering; known mechanisms do it.** The harder question is whether the committed identity is *complete and faithful* — whether it actually represents every transformation, every dynamically generated training example, the authorization state attached to each source, and the resulting checkpoint. **That is an unstandardized engineering and assurance problem rather than an unknown primitive**, and I have not identified it offered as an end-to-end, customer-level assurance at frontier scale.

The other two mechanisms carry no comparable barrier: a corpus commitment and per-customer privacy accounting are engineering problems built from known mechanism classes — a useful commitment needs canonicalization, a manifest definition, preprocessing identity, declared exclusions, versioning and binding to a particular run, and organization-level privacy accounting is not reducible to arithmetic. **Their frontier-scale cost and feasibility are empirical questions.** What they are not is unknown territory, at an accuracy cost I flagged above as one candidate explanation for its absence. I have not found either offered as a standard enterprise product, and their present absence does not reduce cleanly to impossibility.

Which is what makes that wave such a precise piece of evidence. Four organizations have now *committed* — $2.5 billion and 6,000 experts at Microsoft, $1 billion at AWS, more than $4 billion of initial investment in OpenAI's Deployment Company, and a separately funded Anthropic enterprise-services effort — to distributing deployment capability to enterprises, because there is revenue in doing it. Whether it works is not yet knowable; these are commitments weeks old, not delivered outcomes. **What is knowable is what none of them committed anything to.** Deployment, integration and evaluation practice all drew capital. **I have not identified a comparable public capital commitment by a frontier provider centered on independently verifiable, customer-level proof of its own learning-pipeline handling — and it is the only item on that list a vendor would have to point at itself.** I do not read that as conspiracy. I read it as coordination failure — no vendor gains by moving first and no customer can demand it alone — which is the assumption this article flagged, and the reason items 7 and 8 are worth the meeting even when the answer is no.

> **The first frontier vendor to offer a checkable record differentiates on an axis where I have not found a frontier-model provider currently competing.**

One question comes before the four properties: scope

The four properties tell you whether a determination can be checked without trusting its subject. They do not tell you whether the determination addressed the proposition you actually care about. Those are different questions, and the second one comes first.

A perfectly checkable report can still be irrelevant if its declared scope excludes the system, the data path, the learning process, the purpose or the period at issue. **A perimeter-scoped audit does not become evidence about a learning pipeline merely because the audit itself is rigorous.** This is the failure that survives every other control: the determination is honestly made, correctly graded, fully reconstructable, and about something else.

So define the proposition first. Define what has to be inside the assessment for that proposition to be established. *Then* apply the four properties to whatever comes back. Run in the other order and you can spend a procurement cycle establishing, to a very high standard, that a vendor's corporate network is well managed.

And scope has its own authority question, or the gaming simply moves one step earlier. A supplier that can define the proposition narrowly, exclude the difficult learning path before anyone looks, obtain your adoption of that scope, and then never narrow it afterwards will satisfy all four properties honestly. So: **the proposition and the required scope must be defined by the relying party, or by an authority entitled to determine what must be established. The subject may identify exclusions and constraints — it cannot unilaterally determine that its own narrower scope is adequate.**

And this is the honest limit on what follows. The four properties are a specification for the *integrity* of a determination. Scope adequacy is a separate and prior condition, and nothing below supplies it.

Four properties that make a properly scoped determination checkable without trusting its subject

Here they are in full, because each one is doing specific work — because they are what you should require of anybody, from any supplier, before you rely on a record as independently verifiable. The underlying governance principles have familiar antecedents. The contribution here is the way they are assembled into a checkable specification; this article does not characterize the novelty, validity or scope of any pending claim. Each property names a failure that is obvious once stated and invisible until it is.

One. The outcome the record establishes cannot be set by the party it describes. The dispositive outcome **cannot be unilaterally set or overridden by the party being evaluated** — no field, no interface, no admin override that produces a passing result on request. **It may rest on records that party produced**; what must not remain under the subject's unilateral control is the governing rule or the determination it yields. **Without that, the conclusion stays under the control of the party it judges, however official the record looks.** The principle is old and uncontroversial — no organization lets the person who records a transaction be the person who approves it — but it disappears the moment a system reports on itself.

Two. What had to be true is fixed in advance, and the subject cannot narrow it after adoption. The checklist is declared before the work, versioned, and not shrinkable by the subject. **The strong form is authorship by an authority separate from the subject. The minimum acceptable form is that the subject cannot narrow or alter it after adoption** — a supplier may well draft the standard you then accept; what it must not do is shrink it once the result is in view. Without this, "all checks passed" means only that it passed the checks it chose. Fixed *in advance* stops the criteria moving once the result is visible; **non-narrowability after adoption** stops the subject scoping itself into a pass. Separate authorship is the stronger form where the stakes justify it.

And "before" has to be evidenced by something the subject does not control. A timestamp the subject writes does not establish pre-existence — that is the same defect as a self-set verdict, moved onto the clock. **The ordering relation must rest on a time or ordering authority the subject cannot unilaterally set**: an independent timestamping authority, or an append-only log whose consistency somebody else can check.

And where more than one authoritative obligation set applies — negotiated customer obligations, product terms, internal governance rules, statutory constraints — **the rule for precedence, union, conflict and supersession must itself be fixed before the determination and governed under the same authority constraints.** The subject may not choose among competing rosters after seeing the result. Note this is a rule about *who decides*, not a mandate that the most restrictive always wins: an unconditional restrictive-wins rule across thousands of agreements is the anticommons this article warns about elsewhere.

Three. Evidence is graded by what it can actually support; the mapping is fixed before composition under authority the subject cannot unilaterally alter; and classification of a particular artifact into an evidence class is itself governed. A direct observation settles more than a secondhand report, which settles more than an assertion; a thousand weak items do not silently become a strong one, unless their errors are genuinely independent — and independence has to be established rather than assumed. **Without it, a pile of weak evidence adds up to a strong-looking conclusion.** Auditors have long ranked evidence this way — an outside confirmation counts for more than an internal document, which counts for more than management's own account — and the ranking is set by the standard, not by the party being examined.

The last clause of that property is the one people leave out, and it is where the grading actually fails. Fixing a ceiling per evidence class does nothing if the subject can unilaterally alter what the classes mean or what ceiling each carries, and nothing if the subject decides which class a given artifact belongs to. Nobody needs to launder a thousand log lines into an attestation. They need only *call* one an attestation. **Ask who set the mapping, and ask who decides what a given document is.**

Four. The decision behind the record reconstructs from the outside. Somebody who was not there, and who does not trust you, must be able to rebuild it from the package and the evidence it cites — with nothing load-bearing left in unrecorded conversation, private memory, an uncited source, or a limitation dropped while summarizing. **This is the property that converts a record from something you are asked to believe into something a third party can check**, and it is the one most "audit trails" fail.

For consequential determinations that compose multiple sources of evidence, hold a supplier to all four and the properly scoped record can be checked without trusting its subject. Drop one and you may still have evidence — what you no longer have is one of the structural conditions for the form of independent verification defined here. That is a buyer's specification, it is vendor-neutral, and you can hand it to anyone.

One assumption these four properties rest on, stated rather than left implicit. "Evidence" here means an artifact whose **identity, provenance, integrity and custody** satisfy the rules assigned to its evidence class. **These properties govern how a determination is made. They do not turn unauthenticated or mutable evidence into fact**, and a package can satisfy all four while resting on a forged attestation or a broken custody chain. What the properties buy you is that you are relying on declared external trust roots rather than on the subject's say-so — which is why the anchor, the clock and the evidence sources are named later in this article rather than assumed away.

Two things it does not do, stated here rather than discovered later. It does not establish that the determination's scope was adequate — that is the question in the section above, and it comes first. And it does not create any consequence for a false or incomplete record. **A record that is perfectly checkable and carries no liability is a well-made object with nothing behind it**, which is why the clause list earlier in this article spends its capital on the remedy alongside the promise.

Four properties that make a properly scoped determination checkable without trusting its subject

First define the proposition and required scope, under authority the subject cannot unilaterally control. These four then govern the determination, not whether its scope was adequate.

	THE PROPERTY	WHAT IT MEANS	WHAT FAILS WITHOUT IT
1	**The outcome cannot be set by the party the record describes**	no field, no override, no way for the subject to set the outcome on request	the record is the holder's opinion, however official it looks
2	**What had to be true is fixed in advance, and the subject cannot narrow it after adoption**	declared before the work, versioned, and not shrinkable by the subject	“all checks passed” means only that it passed the checks it chose
3	**Evidence is graded by what it can support, before anything is combined**	a ceiling per evidence type, fixed under authority the subject cannot unilaterally alter — and who classifies an artifact into a type is governed too	a pile of weak evidence adds up to a strong-looking conclusion — or a log line is relabelled an attestation
4	**The decision behind the record reconstructs from the outside**	rebuildable by someone who was not there and does not trust you	you take somebody's word for the parts you cannot see

Drop one and you may still have evidence, but not the form of independent verification defined here. Hold all four for a properly scoped proposition and the determination is independently checkable.

Property 2 also requires independent evidence of ordering: the subject cannot establish “before” with a clock it controls.
Where several authoritative obligation sets apply, the rule governing their interaction is fixed in advance too.
Assumption: evidence identity, provenance, integrity and custody satisfy the governing rules for its evidence class.
Vendor-neutral. Applies to any supplier, about anything. Nothing proprietary is needed to require it. Expound Laboratories™ · expound.com

Figure 7 *Four properties that make a properly scoped determination checkable without trusting its subject*

Where I stand in this

I should disclose my position rather than let you discover it. I have spent years building systems that implement exactly those four properties — that is what the next section describes, and it is why I find this problem interesting rather than the other way round. **Every one of the four has a working implementation behind it** — a producer claim, supported at the evidence ceiling set out later in this article: bounded, producer-run evaluation, not independent reproduction or field validation. The specification itself requires none of that: it is stated so a buyer can use it as a procurement criterion without buying anything, and its value is in being adopted rather than owned. The properties are **supplier-independent as evaluation criteria** — which is not the same as supplier-independent implementation, and not a grant of rights to implement protected FAS technology.

> **Licensing and product availability.** The FAS standards family and related protected technology are available for commercial licensing from Expound Laboratories. subject to mutually agreed terms. The public criteria may be used for evaluation and procurement. **Publication of these criteria does not itself grant a license to protected FAS technology.** Separately, Expound currently plans to release its Router soon as open-source software under GNU Affero General Public License v3.0 (AGPLv3). That plan — scope, timing and licensing terms — remains subject to change until release, and rights in any released software will be governed by the license distributed with it. The planned Router release is separate from the FAS standards family and is not intended to release unreleased standards, certification rights, protected marks, confidential know-how or other technology except to the extent actually included in and licensed with the released Router software.

Three further mechanisms harden it, none of them mine, and a serious implementation should use them where the stakes justify the cost: attested execution, so the pipeline produces verifiable evidence of the measured software it ran, for a party who trusts the hardware root rather than the operator; published cryptographic commitments to the corpus behind each released checkpoint; and differential privacy with per-customer accounting, which bounds influence rather than recording it. **Elacity's split-key release has the same architectural property applied to a different object** — a check enforced by a group rather than asserted by an interested one. The property is the right one. I have not identified its application to the governed decision itself in a standard offering.

Adjacent mechanisms already exist, and it would be careless to write as though they did not. Confidential-computing and remote-attestation products for modern accelerators are generally available rather than prospective, and attestation covers multiple accelerators in a single request, with switch-level attestation alongside it. Apple's Private Cloud Compute is the most developed public example of the general architecture: measurements of production images written to an append-only transparency log, images published for independent binary inspection, a research environment for exercising them, and client devices that refuse to talk to a node whose software is not in the log. Zero-retention endpoints narrow contractual and operational exposure by keeping less. **Each of these solves part of the problem described here, and each is stronger than the standard customer-verification mechanisms I have found in ordinary frontier-provider offerings.**

None of them is the object this article is asking for. Confidential computing tells you which software ran on infrastructure you are already trusting to run it. Private Cloud Compute is an inference-time privacy architecture — and, tellingly, even there the published images are not reproducible by outside researchers, so the strongest example in the field still asks for some trust in the builder. Zero retention is a promise about what is kept, which is a contractual object of exactly the kind this article says is not verification. **What I have not found is a standard, customer-verifiable record that a particular customer's contributed experience did not enter an unauthorized shared-learning process.**

> **The check must happen early, and it must be verifiable by someone who does not trust the party holding the record. As of August 2026 I have not identified a standard published enterprise offering from a frontier-model provider that supplies this assurance** — a statement about what I have found, bounded by date and vendor category, which one counterexample corrects. The customer half is built. The vendor half is buildable from components that already exist in adjacent fields and I have not found it offered as a standard enterprise product — which is the point, and the reason the industry's billions are landing on the half that was never the hard part.

What those four properties look like when you build them

Take the specification above and try to implement it, and you run into one obstacle immediately — the same obstacle whether the subject is a training run, a software release, a treatment authorization or a trade.

Organizations decide whether consequential work may be relied on by reading a **status field**: a flag that somebody, or some process, wrote. **A *subject-writable* status field fails property one**, and the precision matters — a field written only by an independent authority, or populated by a governed computation, can satisfy it perfectly well, and a computed verdict has to be serialized somewhere. The defects are **authority** and **staleness**: who may write it, and whether a once-valid determination still holds at the moment somebody relies on it. *Done* is a record of a conclusion someone reached, not a fact about the world, and a cached determination needs a currentness rule. Everything else follows from fixing those two things rather than from avoiding fields.

Finality Assurance™ is the name for what replaces it. It answers one question, and the whole architecture exists to answer it:

> **What may a named party rely upon, about a named result, for a named purpose, now — computed, not written.**

Every clause in that sentence is load-bearing. The answer is specific to who is asking, about what, for what use, at what moment. The same underlying record can return *yes, an analyst may cite this in an internal memo* and *no, a release manager may not route customer traffic on it* — **and neither of those is "the status of the work."** There is no such thing here. That is the first thing most readers have to unlearn.

The mechanics, briefly.

Obligations are declared before the work starts and cannot be narrowed by the subject after adoption, with the "before" resting on a time or ordering authority the subject cannot unilaterally set rather than on a timestamp it wrote itself. Adoption by an authority independent of the subject is the stronger form. That list — the obligation roster — is anchored to a record nobody can revise afterwards, versioned, and not shrinkable by the party being judged. That ordering prevents the failure this whole design is aimed at: *all checks passed, where the caller chose the checks.*

Evidence is typed by what it is capable of establishing, and each type carries a ceiling nothing downstream can lift. A thousand log lines do not become an attestation; a thousand attestations do not become an independent observation; no volume of anything becomes authority. Independence has to be established rather than assumed from organizational separation — three observers watching the same blind spot agree perfectly and know nothing.

The verdict is computed, not written. The result respects a governed ordering over the declared obligations: weaker evidence cannot produce a stronger result, and absent evidence is not treated as a pass. No actor can set it by hand. Where required evidence is simply missing, that is what the record says, and in most workflows *"we checked and it was fine"* and *"nobody checked"* produce the same green tick. Here they cannot.

Verdicts expire. Under **Continuous Finality™**, a verdict is valid at the moment of reliance rather than forever: when the basis for it materially changes, it stops supporting reliance until it has been re-established. **The historical record is never *silently* rewritten.** Corrections, revocations and legally required redactions or deletions are governed transitions carrying their own provenance, so a later reader can tell that the record changed, under whose authority and why. Integrity here means no undisclosed edit — not permanent retention of content the law requires you to delete.

Two more pieces the coda touches. The **Four-Corners Test™** asks whether a decision can be reconstructed from the bounded decision package and the evidence it cites, **with hidden context empty** — no load-bearing dependence on unrecorded chat, private memory, an uncited source, or a limitation dropped in summarizing. It is a sufficiency test, not the verdict. The **Confidence Ladder™** grades maturity across six rungs — defined, exercised, executed, adversarially tested, independently reproduced, field validated — and sets **claim ceilings**: how strongly something may be stated. **A rung is a ceiling, never a permission.** Confidence may inform scrutiny; it may not set finality.

And the piece this article's coda is about. The same components run a governed model router — Governed Multi-Route Selection and Constrained Policy Reinforcement Learning, GMRS and CP-RL. GMRS decides which routes are *admissible* before any optimization happens, and **a route is not a model name**: it includes the evidence plan, the verification path, the stopping rules and the escalation posture, so the same model under a different assurance trajectory is a different route. **That argument is made at length in Today's Routers Pick a Model. The Work Needs a Route.** If nothing is admissible, the correct output is a hold that names the failing condition.

CP-RL is the adaptation layer, and it operates under one constitutional invariant:

> **LearningAuthority ∩ AdmissionAuthority = ∅**

>

> *In words: the authority to learn and the authority to admit share nothing. No overlap, ever.*

The learner may rank routes that were admitted. It may not decide what is admissible, weaken an evidence requirement, broaden a claim license, or promote itself. A learned policy is not entitled to deployment merely because training finished. **Declining to promote a learned policy is the intended behavior, not a failure.**

What that invariant does and does not settle. Disjoint authority prevents the learner from *directly* exercising admission authority, and that is a real structural separation rather than a policy anyone can forget to apply. **It is necessary. It is not by itself sufficient**, and the honest statement of the limit is more useful than the tidy one. Influence can travel indirectly: through the evidence channel, if learner-produced state feeds the predicates that govern its own admissibility; through shared principals, if one person or one service account holds both roles over the same scope; and through route vitality, where a route the learner stops selecting stops generating fresh evidence and can go stale into inadmissibility without anyone deciding anything. **Each of those needs its own control.** Set-theoretic separation of authority is the foundation, not the whole building.

Now map that back to the specification, because the coverage is the useful part.

I have built systems that implement all four of these properties, which is the honest way to present this rather than gesturing at "systems" — and the specification is what matters here, not the architecture behind it. The components are named and described in the **Finality Assurance** technical article; the FAS standards family and related protected technology are available for commercial licensing from Expound Laboratories. **Continuous Finality™** is the related currentness principle: a verdict holds at the moment of reliance rather than forever.

The same evidentiary rule applies here. The verdict is designed as a computation over governed obligations, not as a writable status field. **That is an architectural claim about the system.** Consistent with the standard applied throughout this article, my description establishes the claim I am making, not independent evidence that every deployed instance preserves it. Establishing that requires evidence about the relevant build, execution path and authority configuration.

Four properties, and what to ask a supplier about each one

The first three columns are the specification and are supplier-independent. The fourth turns each property into the question a buyer can put to a supplier, in a questionnaire or a contract.

	THE PROPERTY	WHAT IT MEANS	WHAT FAILS WITHOUT IT	WHAT TO ASK THE SUPPLIER
1	**The outcome cannot be set by the party the record describes**	no field, no override, no way for the subject to set the outcome on request	the record is the holder's opinion, however official it looks	Who can change the result, and can the party the record describes reach that control?
2	**What had to be true is fixed in advance, and the subject cannot narrow it after adoption**	declared before the work, versioned, and not shrinkable by the subject	"all checks passed" means only that it passed the checks it chose	Who fixed the criteria, when, and under whose authority — and can the subject narrow them afterwards?
3	**Evidence is graded by what it can support, before anything is combined**	a ceiling per evidence type, fixed under authority the subject cannot unilaterally alter — and who classifies an artifact into a type is governed too	a pile of weak evidence adds up to a strong-looking conclusion — or a log line is relabelled an attestation	Who decides how much each class of evidence can establish, and who classifies an artifact into a class?
4	**The decision behind the record reconstructs from the outside**	rebuildable by someone who was not there and does not trust you	you take somebody's word for the parts you cannot see	Can an independent reviewer recompute the result from the bounded record and stated rules, without relying on your assertion?

Drop one and you may still have evidence — not the form of independent verification defined here. The questions are the specification put to work; neither needs a particular supplier.

Related currentness principle: Continuous Finality™ — a verdict is valid at the moment of reliance rather than forever, so it stops supporting reliance when its basis materially changes.

Property 2 also requires independent evidence of ordering: the subject cannot establish "before" with a clock it controls.
Where several authoritative obligation sets apply, the rule governing their interaction is fixed in advance too.
Assumption: evidence identity, provenance, integrity and custody satisfy the governing rules for its evidence class.
The protected FAS implementations of these assurance properties are available under license from Expound Laboratories. Publication of these public evaluation criteria does not itself grant a license to protected FAS technology. Expound Laboratories™ · Finality Assurance™

Figure 8 *Four properties, and what to ask a supplier about each one*

The system is designed so that decisions it governs are evaluated under all four properties, and in the bounded evaluations described below those properties were enforced as specified. Where you run it, the record of what was admitted, on what evidence, under whose authority, at what moment, is computed rather than written, graded before composition, and reconstructable by someone who was not there. **That is the customer side of this problem, and it can be implemented on the customer's side without requiring a supplier to change its infrastructure.**

Be precise about what that buys you, because the temptation to overstate it is exactly the failure this article is about. Customer-side governance gives you a reconstructable record of **what you exposed, what you authorized, what evidence supported that decision, and what you relied upon.** It does not prove what the vendor subsequently did with the material. That second proposition lives inside the vendor's infrastructure and needs vendor-side controls or independently anchored evidence.

And there is a subtler point that follows from the specification itself. When an organization runs these properties over its own decisions, that organization is the subject of its own record. Running them makes the decision governed, disciplined and reconstructable *under that organization's own controls* — which is worth a great deal, and is not the same thing as the record becoming independent verification of the organization by itself. **Independence, here as everywhere else in this article, comes from somewhere the subject does not control**: an anchor it does not operate, criteria it did not author, a clock it cannot set, evidence it did not generate. The properties tell you which of those you are missing. They do not supply them.

The vendor side is the vendor's to run. Admission controls over its own learning pipelines, attested execution, a per-customer privacy budget — those are things a model provider does to its own infrastructure, **and here the taxonomy has to be exact rather than tidy. Most of the vendor-side control plane is buildable from components that already exist. A bounded assurance question remains: whether the committed corpus and authorization identity completely and faithfully represent what the learning process actually consumed through preprocessing, dynamically generated material, execution and the released checkpoint. That residue does not stop a buyer contracting for outcome and control obligations today — which makes this predominantly a market and procurement vacancy containing a bounded assurance residue.**

The scarce thing here has never been implementation. It is agreement on what a trustworthy record has to be, stated well enough that a buyer can require it and a supplier can be measured against it. The four properties are that statement. What I have built implements the half that sits on your side of the line — and the half that does not is now a specification somebody can be held to rather than a wish.

Each component above is described in the Finality Assurance™ core technical paper at the level of what it does. Implementing mechanisms are held as trade secrets and are not disclosed there or here. The notice at the end of this article carries the full IP and status position.

Where I came from, and what I actually observed

A short coda, because it explains why I care rather than what you should do.

I did not arrive here from economics. I arrived from that operational problem, and it produced a sequence of questions, each forced on me by the previous answer failing. The machinery in the section above is what each answer turned into.

What establishes that a model's output may be relied on? Not the output's confidence. A system stating something firmly does not establish that it may be relied upon — which is why the verdict is computed from a roster declared in advance rather than asserted by the thing being judged.

Here is what that answer does to work that should never have shipped. Across **2,520 graded answers from six language models on verifiable tasks, paired task for task — 1,260 per arm** — the ungoverned side, which ships on the model's own claim of completion, shipped **seven wrong or fabricated answers as finished work: 7 of 1,260, about 0.56%.** The governed side shipped **none: 0 of 1,260. That is the comparison that matters, because shipping on the model's own claim of completion is a common deployment pattern in current agentic systems** — and it is exactly the condition governed finality replaces. On a separate and much smaller battery, a smaller frontier model omitted a required interface contract — a declared input and output guarantee — and reported success in five attempts out of five; governed finality shipped that broken work **zero times out of five.**

The validation posture, stated here rather than left to be discovered. These are bounded, producer-run evaluations. The protocols were fixed before execution and retained with the evaluation estate, and the harness that runs them is available to any buyer who asks under agreement. They establish performance on the stated batteries; **they are not independent field validation, and independent reproduction is a separate threshold that has not been crossed.**

And here the specification this article asks of vendors applies to me, so let me be exact about where my own results sit under it. The obligation rosters were declared and fixed before any run and were not narrowed afterwards — property two in its *minimum* form, since I authored them; separate authorship is the stronger form and I have not met it. Evidence is typed, with ceilings fixed before anything was combined. Every disposition is a computed verdict. **Property four is the one where the gap is real**: a harness offered to buyers under agreement is an *offer* of reconstruction, not a reconstruction by somebody who owes me nothing. On the six-rung ladder above, this evidence supports *adversarially tested*. It does not reach *independently reproduced*, and I should not write as though it did. **A specification its author will not apply to his own results is a marketing document** — and applying it honestly costs a claim I would rather have made.

And it does that without simply refusing everything, which is the failure mode that makes strict gates useless. On a 30-task legitimate-work battery run nine times over, **zero of 270** correct, evidenced answers were refused. On a 720-trial hazard campaign — 360 planted hazards against **360 matched clean trials** — the result was **zero permits on the planted-hazard arm and zero false**

refusals on the matched clean arm. The matched arm is the part that matters: catching everything is trivial if you stop everything, and the second half is what makes the first half mean anything. Alongside that: **46 of 46** deliberately broken versions caught before release, and **17 of 17** negative controls fired — a negative control being a test rigged to fail, run to prove the checker is capable of failing at all.

What establishes which way of doing the work may be used? Not capability. A system can be entirely capable of a task and still not be an admissible route for it — which is the question the governed router exists to answer, one route at a time.

And the router carries the same discipline into cost, by a mechanism worth naming rather than leaving to inference. On a frozen 36-task battery, governed routing matched a static frontier comparator exactly — **36 of 36 correct dispositions, 18 of 18 safe declines, zero false accepts on both arms** — while using **24 model calls rather than 36, and no frontier calls at all. The saving is not a mystery and is not an optimizer**: twelve of the thirty-six tasks were routed to exact deterministic computation, which is where 36 minus 12 equals 24 comes from. Deciding that a third of the work did not need a language model at all *is* the result. Both arms scoring perfectly on dispositions means this battery separates the two on cost rather than on quality, and that is what it was built to do. A live smoke campaign at the same size reproduced it: identical disposition, 24 calls against 36, **4,042 tokens against 6,849**. Widen to a 240-task live campaign and calls, tokens and cost all fall again — 199 against 240, 122,165 tokens against 152,075, lower on both modeled and observed cost.

And the reason those numbers are interpretable under the stated protocol is the ordering. Disposition and false acceptance are measured *beside* cost, not after it. A cheaper route that quietly loses quality is identifiable as one before anything ships — where an ungoverned optimizer would report the cost line, which always looks excellent, and nothing else. **Efficiency does not get a vote on the constraints.**

What establishes which improvements a policy may make to itself? Not the improvement. A policy winning on its assigned objective can be losing on the outcome you care about, and the summary figures will not tell you. So here the architecture supplies something structural rather than a measured rate: the authority to learn and the authority to admit are held disjoint.

That is the invariant stated in full in the machinery section above. The learner may rank routes that were admitted. It does not directly possess the authority to admit a new one, weaken an evidence requirement, broaden a claim license, or promote its own updated policy — promotion is a verdict issued from outside the learner, on evidence the learner does not control. **It does not, by itself, prove complete non-interference**: indirect influence through the evidence channel, shared principals, route vitality or the surrounding control paths needs its own controls, as set out above.

A measured refusal rate tells you how often a learner was stopped. Authority separation tells you something different: the learner was never granted that direct authority in the first place — and unlike a rate, that does not degrade as the learner gets better at its job. Neither is the whole answer, and the two are worth having together. **A learner cannot be allowed to define its own success.**

And one more, which is where this article started. What establishes that a system permitted to *answer* is permitted to *learn*? Not the permission to answer. **A system being capable of learning from an interaction does not establish that it was entitled to** — which is the same shape as the question this whole article is about.

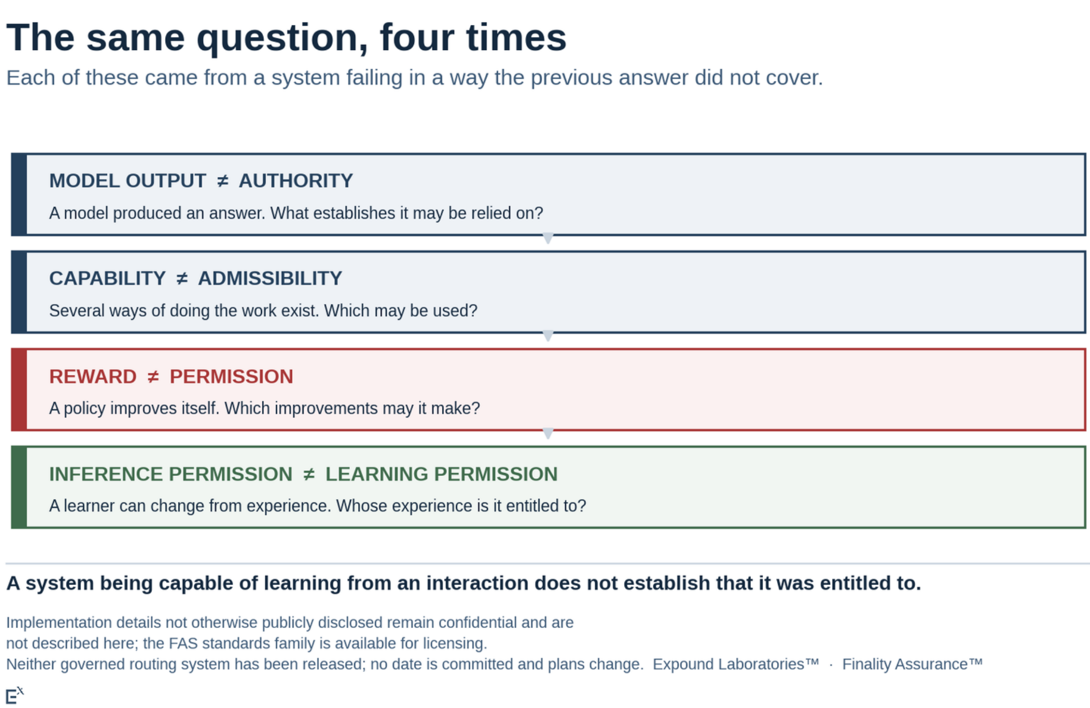

Figure 6 *The same question, four times*

Key takeaways

1. **You certainly hand over two things — money and knowledge — and only one is invoiced.** Whether the second is being collected as a payment is unestablished, by anybody, in either direction. That is the finding, not a hedge.
2. **"Inference changes nothing" is true of the model and misleading about the system, and "we do not train on your data" is not the covenant you want.** The weights do not move when you ask a question. Serving visibly writes memory records and index entries; vendors describe feedback events and traces on top of that. That durable non-parametric layer is the least governed — in many enterprise arrangements broader vendor use depends on specified exceptions, and those exceptions sit in a carve-out that is easy to overlook.
3. **Arrow explains the disclosure asymmetry at purchase. What follows is holdup, plus a spillover risk that cannot be ruled out.** Holdup has known contractual remedies. It is the *possibility* of spillover that makes verification, rather than a remedy against your counterparty alone, the load-bearing requirement.
4. **Trade secret law already reaches misuse without requiring you to prove copying** — except where your own contract already licensed the use, which for the product-improvement carve-out is present in every enterprise agreement in the sample I reviewed, **a complete statement about my sample and none at all about yours. Those exceptions matter twice: they create channels for secondary processing, and they may later give the vendor a contractual argument that the challenged use was authorized.** Trade-secret law is generally the closer fit for the confidential operational know-how at issue, and the patent analogy points away from it.
5. **FTC orders have required deletion of models, algorithms or derived work product, without solving attribution** — so "nothing can be done afterwards" is too strong. It is untested as a compelled remedy, and **that specific FTC instrument is not one a private enterprise invokes for itself** — what private relief would be available in your own dispute is a separate question these matters do not settle.
6. **Attribution is not impossible.** It is **generally too noisy and contestable to carry a claim by itself** — though it can contribute alongside access, timing, similarity and discovery. A weaker claim, and a true one.
7. **The standards have no machine-readable word for it.** As of revision -07 the IETF vocabulary defines only model training and search, and its search category requires linking back to the source and excludes summarization — so enterprise retrieval falls between the two, and the draft takes no position on what silence implies. CAWG does distinguish generative from non-

generative training; its limit is that all four entries attach to a *file*, not to a trace, an eval or a memory record.

8. **A vendor-generated log may well be evidence. It does not become independent verification because it is detailed, signed or timestamped.** That takes someone who is not the vendor.
9. **Differential privacy with per-customer accounting is the strongest formal guarantee discussed here for bounding contribution-level parametric influence, where you *authorize* learning** — not for memory or retrieval, at a material utility cost that is mechanism and task dependent, only as good as the unverified claim that it was run, and **it bounds how far a trained model's behavior can differ depending on whether any single contributor's data was included, rather than establishing that your material was never used.** If what you need is no learning at all, it solves the wrong problem. Ask for the privacy unit and the accounting method, not the label: record-level, user-level and organization-level guarantees are different things. I have not seen it routinely requested in the agreements and procurement materials I have reviewed. Whether that is *why* it is not offered is the assumption this article flags, not a demonstrated fact.
10. **Two gaps remain, not one:** the unstandardized entitlement over the durable non-parametric layer, and no affordable independent verification I have been able to identify in a standard frontier-vendor offering as of August 2026, which is a verification gap — an engineering and procurement problem rather than a legislative one. The second is larger and is the one you can act on without waiting for anybody to legislate.
11. **Capability is being distributed. Verification is not.** Microsoft committed $2.5 billion and 6,000 experts to building the loop inside enterprises; AWS, OpenAI and Anthropic funded equivalents within weeks. That investment is real, and deployment capability is being commercialized and distributed to firms that could never have staffed it. **Whether the newly announced programs deliver the promised outcomes is too early to establish** — they are commitments weeks old, not delivered results. **The capability I have not found a comparable public commitment to is independently verifiable, customer-level proof of what the supplier did with what you gave it, and it is the only item on the list a vendor would have to point at itself.** Half of that is yours to fix unilaterally today; the other half needs the model vendor, and is therefore a market and procurement vacancy containing a bounded assurance residue.
12. **The clause list is the deliverable, and its first item is a reframing.** Negotiate a *no shared learning* covenant rather than a *no training* one: your material can alter a shared service through routing, evaluation, distillation, safety models, retrieval corpora or exemplar construction without meeting anybody's definition of training. A negotiated scope is enforceable, and in the agreements I have reviewed buyers routinely leave it unexpressed — which is the cheapest work in this article and the only part that works without anybody building anything new.

13. **Any scheme of this kind has one hard constraint.** A veto over how a lawfully held input may be used is exclusionary **in the economically relevant sense — you retain a veto over a defined use, which does not require treating the knowledge, the model or the derived state as your property** — and reconciling one across thousands of agreements would gate any shared learning process on its most restrictive customer unless exercising it costs machine time rather than legal time. That is the case for declaring obligations in a machine-evaluable form rather than in prose — though note that the clause list above, which is this article's practical deliverable, is prose negotiated in legal time, and the constraint bites on the scheme rather than on the negotiation.
14. **Almost all of it reduces to friction. The durable non-parametric layer has the least standardized entitlement structure.** Particular artifacts and uses may already be covered by contract, confidentiality, trade-secret, privacy or other law — but no general customer exclusion rule spans the category. And the one bar you created yourself is the carve-out you signed — and even that is a clause you can renegotiate. **Most of the rest is priced out of reach rather than forbidden** — with one exception the article keeps rather than smooths over, and it is narrower than it first appears: the residue is **not attesting execution**, which is a shipping mechanism, and not binding a manifest to it, which is known engineering — it is **establishing that the committed corpus and authorization identity faithfully represent what the learning process actually consumed**. The contracting and admission-control arguments here do not depend on that binding being solved. **Friction is what makes a right safe to ignore, and it is the one barrier that falls without permission from anybody.**

Four questions to put to your own organization

- **If a supplier's own engineers built our AI deployment, who is left to verify it** — and did we budget for anyone independent of them?
- **Who inside this company has actually read the product-improvement carve-out in our largest AI agreement**, and when?
- **If we learned tomorrow that our corrections had reached a competitor, what would we do differently** — and does that answer justify what we are currently spending to find out?
- **If our own systems learn, what stops the learning from deciding what counts as success?**

My position

Nadella asked a good question. Arrow explains the purchase-side information problem; it does not explain the relationship-specific investment that follows. Nadella's own answer addresses that second problem with the most expensive instrument available. The answer therefore need not begin with a new property right. In many enterprise settings relevant protections may be available through contract, trade-secret law, data protection law or sector-specific requirements — and, for the parametric channel, for anyone willing to ask, in a mathematical bound rather than an assurance. Whether they actually apply, and what they permit or prohibit, depends on the jurisdiction, the facts, the data, the parties' legal roles, the configuration and the agreement. The larger practical gap is making the relevant restriction reach the real learning path, and making compliance independently checkable.

What is not yet standard — not absent, not unbuilt, but not on offer from the party you buy from — is a way to check cheaply, without litigation, by someone who does not have to trust the vendor.

So return to the ledger this article opened with. The hypothesis — that the second payment is being collected — is exactly where it was: unresolved, and not resolvable from where you are standing. I have not moved it, and I do not think anyone writing about this subject has. **What has moved is the diagnosis of why it is stuck** — and one half of that diagnosis is firmer than the other. That it is *not* stuck because the law is silent, because the standards failed, or because attribution is impossible, I think this article establishes: those are checkable claims about published instruments and available methods. **That what remains is a coordination failure rather than an absence of demand is the part I have argued and not demonstrated**, and it is still the part I would most like to be shown wrong about. What I will stand behind is the shape of it: every route to an answer costs more than the answer is currently worth to the only party who would pay for it.

Which returns this to where it started, and to where the two articles before it ended. The obstacle is not the absence of a legal basis for asking, or for contracting for protection. It is cost — the cost of reading the clause, of drafting the term nobody standardized, of running an attribution study that remains contestable in litigation, of asking for an assurance I have not found routinely requested in current procurement practice. **Most of those obstacles are prices, and prices move.** One bounded assurance residue remains — ensuring that the committed corpus and authorization identity completely and faithfully represent what the learning process actually consumed — and it does not stop an enterprise contracting today for the controls and outcomes that are already technically enforceable.

> **The legal ability to restrict much of what your vendor may do is not the missing piece. The affordable ability to determine whether the restriction was respected is — and that is a price rather than a prohibition — which is the same sentence I have now written about patents, about enforcement, and about this. Friction is the actionable part of the problem, and that is the good news: a price moves when somebody builds a cheaper instrument, and nobody needs permission to do it.**

So settle the ledger. Whether the second payment is real remains open, and I have not identified a standard offering that closes the chain end to end at the customer-level granularity described here. **What I have not found is routine independent verification at the customer-level granularity described here, in any standard frontier-vendor enterprise offering as of August 2026** — and of the two facts, it is by far the more actionable, because it is the one with an instrument attached.

And the reason to say so plainly is that friction is the good news. A missing right waits on a legislature. A price waits on somebody building the cheaper instrument — and on enough buyers asking that building it pays. The first of those is out of your hands. The second is a line in your next procurement questionnaire.

> **The next AI governance problem is not how to control what models say. It is how to govern what they are permitted to become — and to make confirming it cheap enough that somebody other than the vendor actually does.**

Sources

Quotations from the Nadella essay were verified against Nadella's *author-hosted July 12, 2026 copy at sn scratchpad* and cross-checked against his *original X publication*, not from the summaries or reposts, several of which are translations that misrender his terms.

Satya Nadella, *The Reverse Information Paradox*, sn scratchpad, July 12, 2026 · Arrow, *Economic Welfare and the Allocation of Resources for Invention*, NBER/Princeton 1962 · Demsetz, *Toward a Theory of Property Rights*, 57 Am. Econ. Rev. 347 (1967) · Heller, *The Tragedy of the Anticommons*, 111 Harv. L. Rev. 621 (1998) · Williamson, *Markets and Hierarchies* (1975) and *The Economic Institutions of Capitalism* (1985); Klein, Crawford & Alchian, *Vertical Integration, Appropriable Rents, and the Competitive Contracting Process*, 21 J.L. & Econ. 297 (1978).

Defend Trade Secrets Act, 18 U.S.C. §1836 et seq. · *Trinidad v. OpenAI, Inc.*, No. 4:25-cv-06328-JST (N.D. Cal.) (Tigar, J.) (pro se; dismissed at the pleading stage for failure to plead reasonable measures; Ninth Circuit appeal No. 26-721 dismissed Apr. 6, 2026 under 9th Cir. R. 42-1, the district court having certified that the appeal was not taken in good faith) · FTC algorithmic disgorgement matters: *Cambridge Analytica* (2019), *Everalbum/Paravision* (2021), *Kurbo/WW* (2022), *Edmodo* (2023) and *Ring* (2023) — four resolved through settlement, consent or stipulated orders, *Cambridge Analytica* by administrative default; none litigated to judgment on contested merits. The *Edmodo* order requires deleting "models or algorithms developed using personal information collected from children" without the required consent, and *Ring* requires deletion of data products including models and algorithms. The contemporaneous *Amazon Alexa* order is deliberately excluded from this list: it requires deletion of covered data and bars its use to create or improve data products, which is a different remedy · EDPB Opinion 28/2024 on personal data in AI models.

CAWG *Training and Data Mining* assertion v1.1, DIF-ratified May 16, 2025, §3.1 (four entries: `cawg.data_mining`, `cawg.ai_inference`, `cawg.ai_training`, `cawg.ai_generative_training`) — cawg.io/training-and-data-mining/1.1 · IETF *draft-ietf-aipref-vocab-07*, published August 19, 2026, §§3.2, 4.1, 4.2, 5, 5.1, 5.2 — note the draft's own statement that its contents "DO NOT REFLECT CONSENSUS of the Working Group either in whole or part." The frequently quoted line "preferences do not themselves create rights or prohibitions" appears in revisions -03 through -05 and was removed in -06; it is not in the current text.

Elacity Labs · NeoSapients · Marisetty and Mamidi, SSRN 7115525.

Adjacent mechanisms. Apple Security Research, *Private Cloud Compute: A new frontier for AI privacy in the cloud* and *Security research on Private Cloud Compute* — transparency log of production image measurements, images published for binary inspection, and the Virtual Research Environment. Apple publishes production binaries, offers transparency-log verification, and makes only a subset of source available under a limited-use license; the verification it offers is binary-against-log rather than source-to-binary reproduction. I therefore do not treat the mechanism as independent build reproducibility — that conclusion is my inference, not Apple's stated claim. NVIDIA, *Attestation* — GPU attestation and nvSwitch attestation; multi-GPU (PPCIE) attestation is documented separately and linked from that page (page last updated August 1, 2026); *NVIDIA H100 GPUs in Microsoft Azure confidential virtual machines, general availability*.

Vendor documentation cited in the clause section, each checked against the published page rather than a summary, **as of August 2026 — vendor terms and feature defaults change frequently, and every one of these should be re-verified against the current page before you contract on it**: Google, *Vertex AI and zero data retention*, Google Cloud documentation — training restriction under Service Specific Terms §17, and the feature-by-feature retention table including the thirty-day Grounding storage that cannot be disabled and the default in-memory caching of inputs, outputs and derived data. Anthropic Privacy Center, *Is my data used for model training?* — commercial inputs and outputs not used for training by default, explicitly submitted feedback may be, related conversation retained up to five years, and a Team and Enterprise setting to disable feedback. Microsoft Learn, *Data, privacy, and security for Models sold by Azure* — models stateless at the model level, with the Responses API, Assistants threads and stored completions persisting message history, and abuse monitoring able to select a sample of prompts and completions for human review.

By the same author. *Who Can Afford Friction?* · *Startup IP Is the Most Underpriced Asset on Your Cap Table* · *Today's Routers Pick a Model. The Work Needs a Route.* · the *Finality Assurance™ core technical paper*, which describes the components named above at the level of what each does.

On motive: Microsoft Frontier Company was announced July 2, 2026 — $2.5 billion and 6,000 industry and engineering experts, with Judson Althoff's "beyond what has been labeled as Forward-Deployed Engineering" and named early engagements at the London Stock Exchange Group, Unilever and Land O'Lakes, with Accenture named as a delivery partner (TechCrunch, CNBC, July 2, 2026). AWS announced a $1 billion equivalent on June 30, 2026; OpenAI and Anthropic launched comparable ventures in May 2026. The identification of Copilot and Azure AI Foundry as Microsoft's own answer to the problem the essay describes was given by a Microsoft spokesperson to *The Register*. Luis Garicano (London School of Economics) characterized the essay as "smart econ thinking." Headlines quoted from *The Register* and *The Next Web*, July 2026. Everything else in that section is labeled inference and should be read as mine rather than as reporting.

This article describes the author's research, experience and operating posture. It is not legal advice, and nothing here should be relied on as a substitute for counsel on your own contracts. Finality Assurance™, Continuous Finality™, Semantic Computation™, the Four-Corners Test™, the Confidence Ladder™ and Expound Laboratories™ are claimed marks of Kenneth L. Cheney; where a trademark application has been filed it is pending examination, no mark is registered, and no claim of federal registration is made or implied. Patent applications covering the Four-Corners Test™ and the Finality Assurance™ verdict mechanism are pending; no patent has issued, and no claim of an issued patent is made or implied. Implementing mechanisms are maintained as trade secrets and are not disclosed here. Governed Multi-Route Selection, Constrained Policy Reinforcement Learning, Finality Verdict and Finality Account are technical names used descriptively and are not claimed as marks; neither system has been released, no release date is committed, and plans *change. Third-party names are used descriptively and nominatively; no affiliation, endorsement or partnership is implied.*

About Expound Laboratories

Meaning, made computable. Finality, made governable.

Expound Laboratories develops research, standards and computational systems concerned with **Semantic Computation™** and **Finality Assurance**. The work is motivated by a distinction that becomes more consequential as computational systems become more capable: **producing an output is not the same as establishing what that output means, and neither is the same as determining whether it has earned a defined status for reliance.** Three questions that ordinary workflows collapse into one:

> **What was produced? ≠ What does it establish? ≠ Under what conditions may it be relied upon?**

Semantic Computation concerns the explicit representation and computation of meaning-bearing structures — claims, evidence, context, requirements, obligations, dependencies, authority, policy, provenance, contradictions, permitted transitions. **Finality Assurance** addresses the separate question of when work has satisfied the applicable evidence, policy, authority, scope and temporal conditions for a governed disposition. The first develops the semantic state; the second governs its disposition. **A refusal is a verdict, and issuing one is not the same as satisfying it.**

> **Finality Verdict issued ≠ Finality satisfied**

The mark is the natural exponential, e^x, read two ways. Mathematically, *e* is Euler's number and the function has the property that its rate of change equals its current value — a compact object implying a much richer structure of direct and indirect relationships, which is the representational analogy intended. Semantically, the *e* stands for Expound and the *x* for the unresolved subject: a question, a claim, a decision, a requirement, a system state. Read as a progression, the mark says **expound the unknown, compute the meaning, govern the finality.**

The analogy is representational and is not a claim about implementation. The exponential also carries the limitation the architecture exists to address: **an exponential has no notion of completion, sufficiency, authority or reliance, and a system's ability to keep generating does not establish that its output is finished.**

> **Expansion ≠ Finality · Scale ≠ Correctness · Confidence ≠ Authority**

Expound means to unfold and make meaning explicit; **Laboratories** adds the requirement to test it — hypotheses, controlled computation, evidence and provenance, measurement, adversarial testing, reproducibility, falsification, refusal. The combination is deliberate in both directions. **Semantic elaboration without experimental discipline amplifies a wrong premise, and a closure mechanism operating over underspecified semantics authorizes a result whose meaning was never established.** Finality here is not permanent or context-free certainty; it is an earned status inside an explicitly bounded jurisdiction of reliance — which is why every result is reported with the estate that produced it.

What to do with this

Evaluate. Use the publicly stated assurance properties to examine what your current vendor and contract actually establish. You can do that without taking my implementation claims on faith — that independence is the point of stating them separately from any supplier.

License. If you want the protected FAS standards family and related technology rather than reconstructing it from public descriptions, Expound Laboratories licenses it. Publication of these criteria does not itself grant a license to protected FAS technology.

Watch for the Router. Expound currently plans to release its Router soon as open-source software under AGPLv3. Scope, timing and licensing terms remain subject to change until release, rights in any released software will be governed by the license distributed with it, and the planned Router release does not open-source the FAS standards family, certification rights, protected marks, unreleased implementations or confidential know-how.

We are speaking with enterprises, model providers, researchers and infrastructure partners interested in evaluation, licensing and deployment.

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